

Java Language

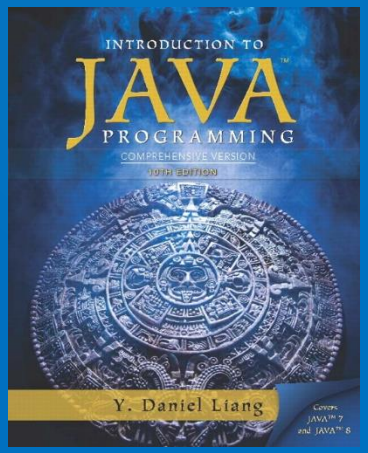
Wang Xiaomeng

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College of Computer & Information Science

Book:

Java语言程序设计（基础篇）（英文版·第10版），[美] 梁勇（Y.Daniel Liang）著.



Reference:

1. Think in Java, Fourth Edition. 2006.

2. http://docs.oracle.com/javase/8/docs/index.html

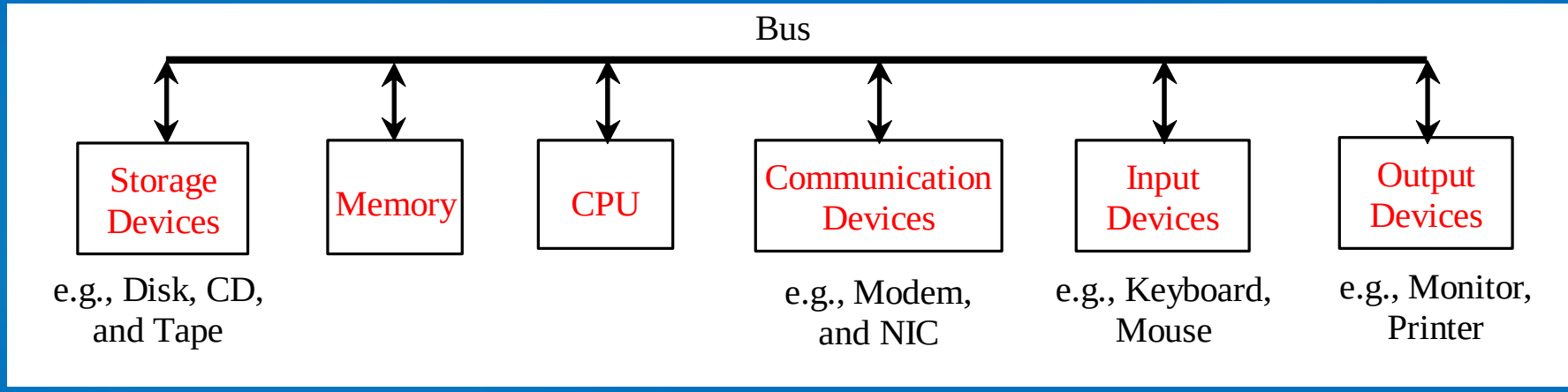
3. http://docs.oracle.com/javase/8/docs/api/

4. https://docs.oracle.com/javase/specs/jls/se8/html/index.html

1 Introduction to Java

What is a Computer?

A computer consists of a ***CPU***, ***memory***, ***hard disk***, ***monitor***, ***printer***, ***communication devices***, et al.



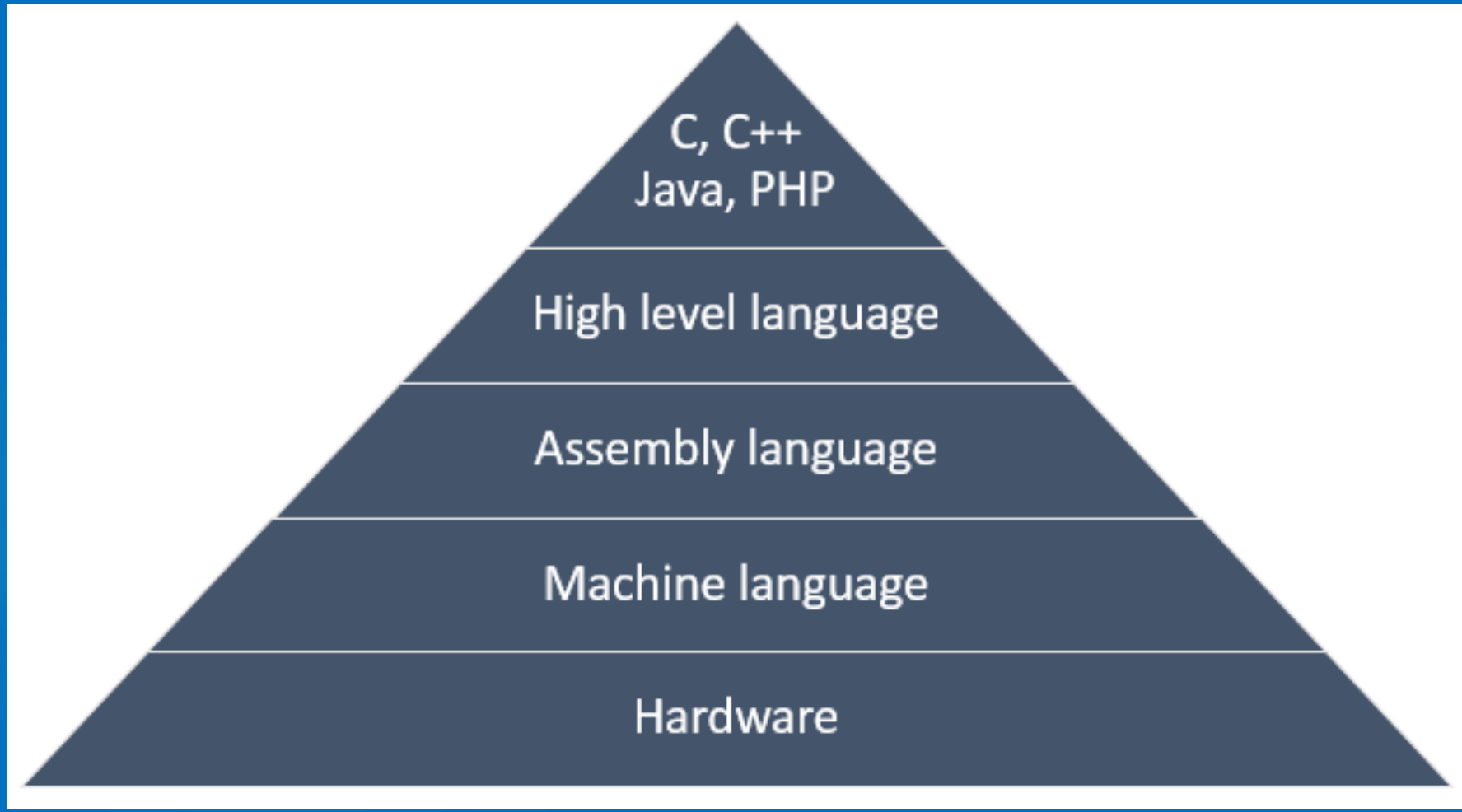
Programs

Computer programs, known as *software*, are *instructions* to the computer. You *tell* a computer what to do *through programs*. Without programs, a computer is an *empty* machine. Computers do not *understand* human *languages*, so you need to use computer languages to *communicate* with them. Programs are written using programming languages.





Programming Languages



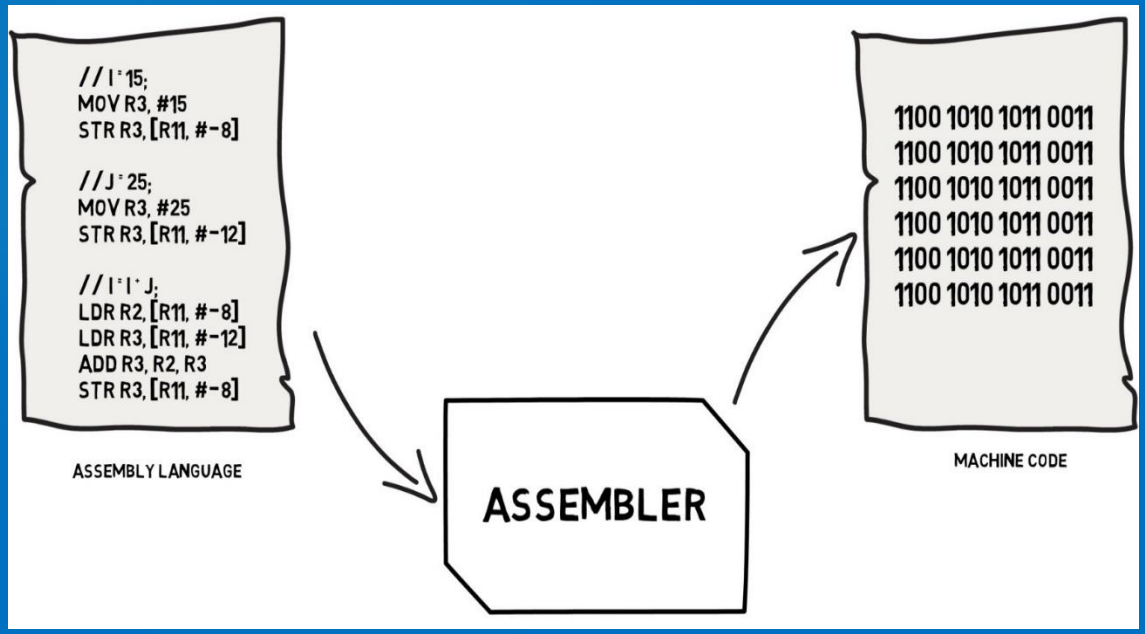
Machine language

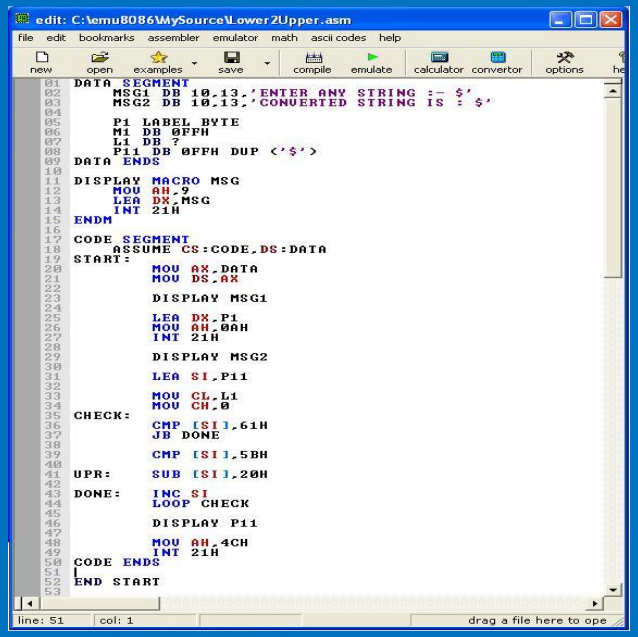
Machine language is a set of *primitive instructions* built into every computer. The instructions are in the form of *binary code*, so you have to enter binary codes for various instructions. Program with native machine language is a *tedious* process. Moreover the programs are highly *difficult* to *read* and *modify*. For example, to *add two numbers*, you might write an instruction in binary like this: **1101101010011010**



Assembly languages

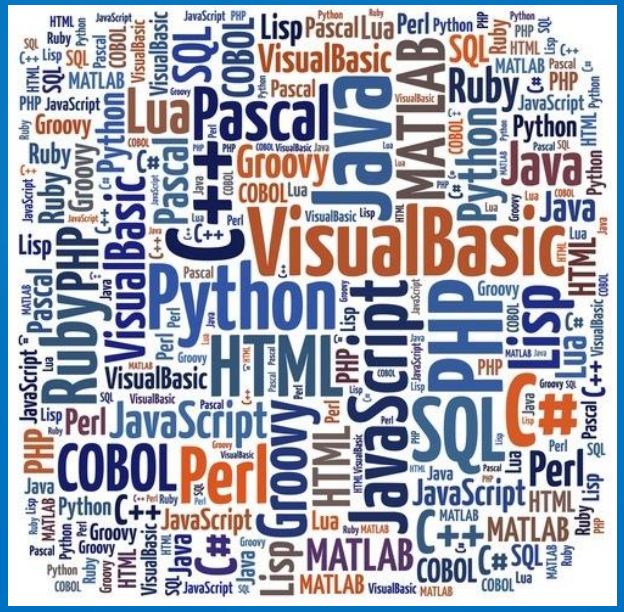
Assembly languages were developed to make *programming easy*. Since the computer cannot understand assembly language, however, a program called *assembler* is used to *convert* assembly language programs into machine code. For example, to add two numbers, you might write an instruction in assembly code like this: ADDF3 R1, R2, R3





High-level languages

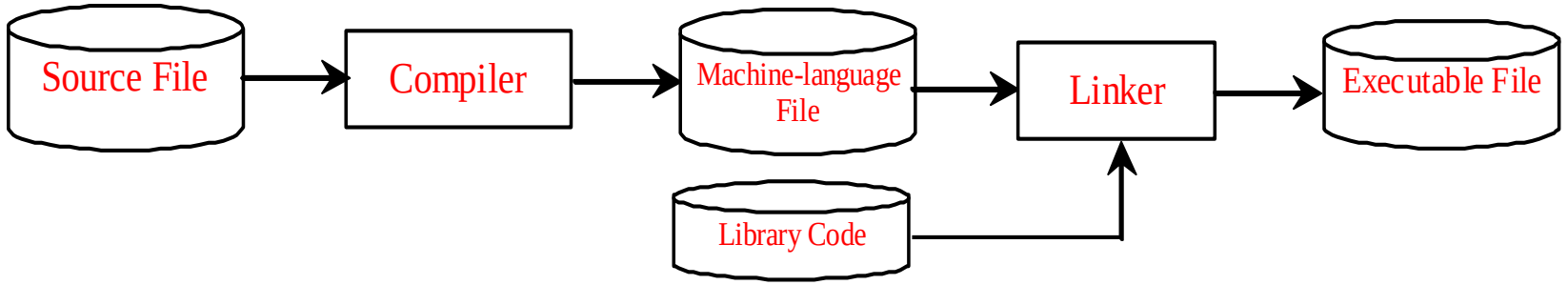
The *high-level languages* are *English-like* and easy to learn and program. For example, the following is a high-level language statement that computes the area of a circle with radius 5: *area = 5 * 5 * 3.1415;*




```php
<?php
    function answerDistribution($dbconn, $date) {
        $result = mysqli_query($dbconn, "SELECT * FROM TechTerms WHERE Date='$date'");
        if($result !== false) {
            $distArray = array();
            $row = mysqli_fetch_assoc($result);
            $correctAnswer = $row['Correct'];
            $distArray['A'] = $row['Anum'];
            $distArray['B'] = $row['Bnum'];
            $distArray['C'] = $row['Cnum'];
            $distArray['D'] = $row['Dnum'];
            $distArray['Correct'] = $correctAnswer;
            $distArray['Answer'] = rtrim($row[$correctAnswer], ".");
            $distArray['Query'] = "SELECT * FROM TechTerms WHERE Date='$date'";
            return $distArray;
        } else {
            $distArray['Error'] = 'Quiz load query failed';
            return $distArray;
        }
```

Compiling Source Code

A program written in a high-level language is called a *source program*. Since a computer cannot understand a source program. Program called a *compiler* is used to *translate* the source program into a machine language program called an *object program*. The object program is often then *linked* with other *supporting library* code before the object can be executed on the machine.



Popular Languages

https://www.tiobe.com/tiobe-index/

Aug 2020	Aug 2019	Change	Programming Language	Ratings	Change
1	2	↑	C	16.98%	+1.83%
2	1	↓	Java	14.43%	-1.60%
3	3		Python	9.69%	-0.33%
4	4		C++	6.84%	+0.78%
5	5		C#	4.68%	+0.83%
6	6		Visual Basic	4.66%	+0.97%
7	7		JavaScript	2.87%	+0.62%
8	20	⇈	R	2.79%	+1.97%
9	8	↓	PHP	2.24%	+0.17%
10	10		SQL	1.46%	-0.17%
11	17	⇈	Go	1.43%	+0.45%
12	18	⇈	Swift	1.42%	+0.53%
13	19	⇈	Perl	1.11%	+0.25%
14	15	↑	Assembly language	1.04%	-0.07%
15	11	⇊	Ruby	1.03%	-0.28%

Why Java?

The answer is that Java enables users to develop and deploy applications on the *Internet* for *servers*, *desktop* computers, and *small hand-held* devices.

- Java is a *general purpose* programming language.
- Java is the *Internet* programming language.



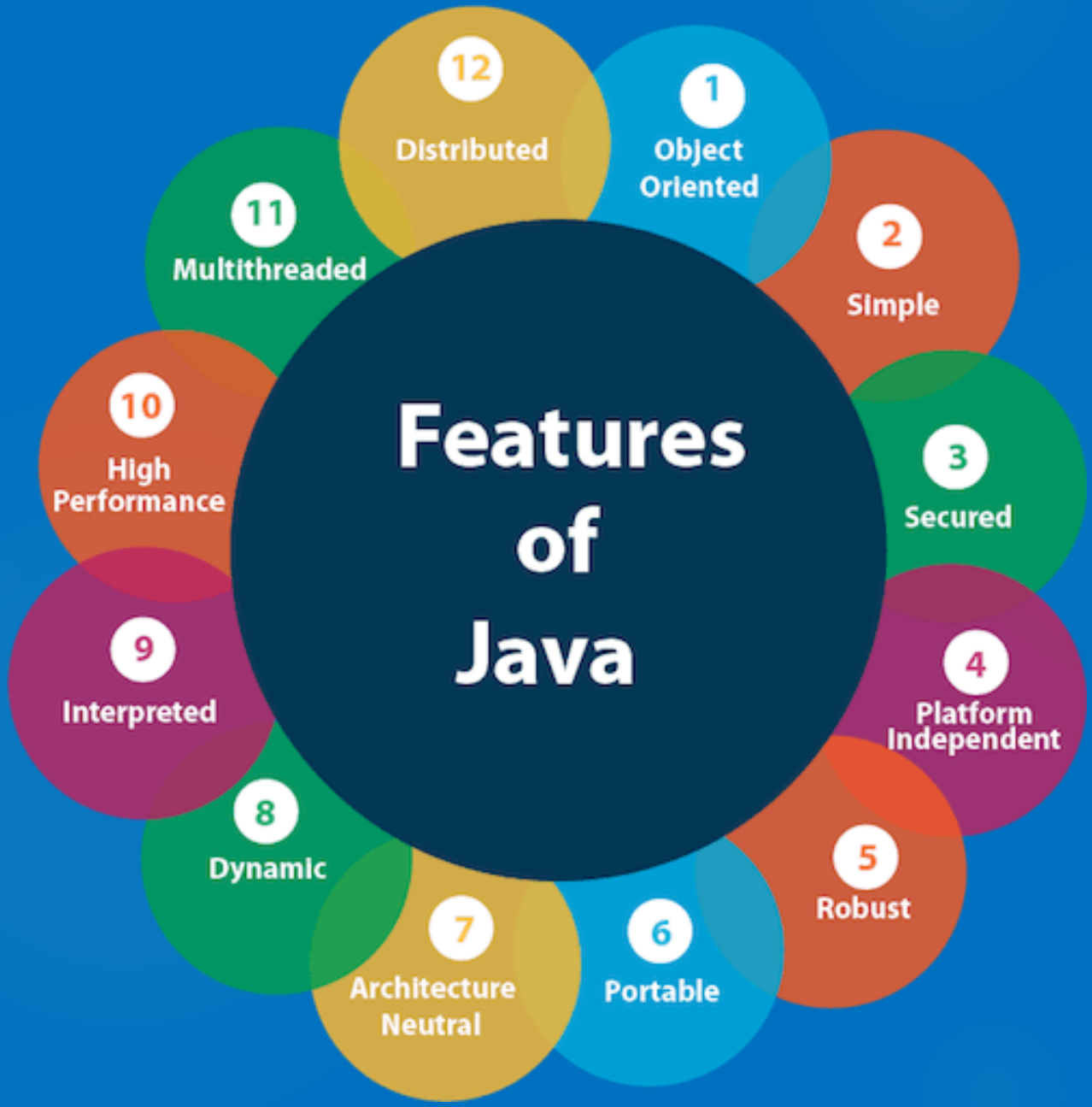
History of Java

- *James Gosling* and Sun Microsystems
- Oak (1992)
- Java, May 20, 1995, Sun World
- HotJava
  - The first Java-enabled Web browser
- 2009, Oracle and Java
- Latest version: Java SE 14








Features of Java
1 Object Oriented
2 Simple
3 Secured
4 Platform Independent
5 Robust
6 Portable
7 Architecture Neutral
8 Dynamic
9 Interpreted
10 High Performance
11 Multithreaded
12 Distributed

Simple

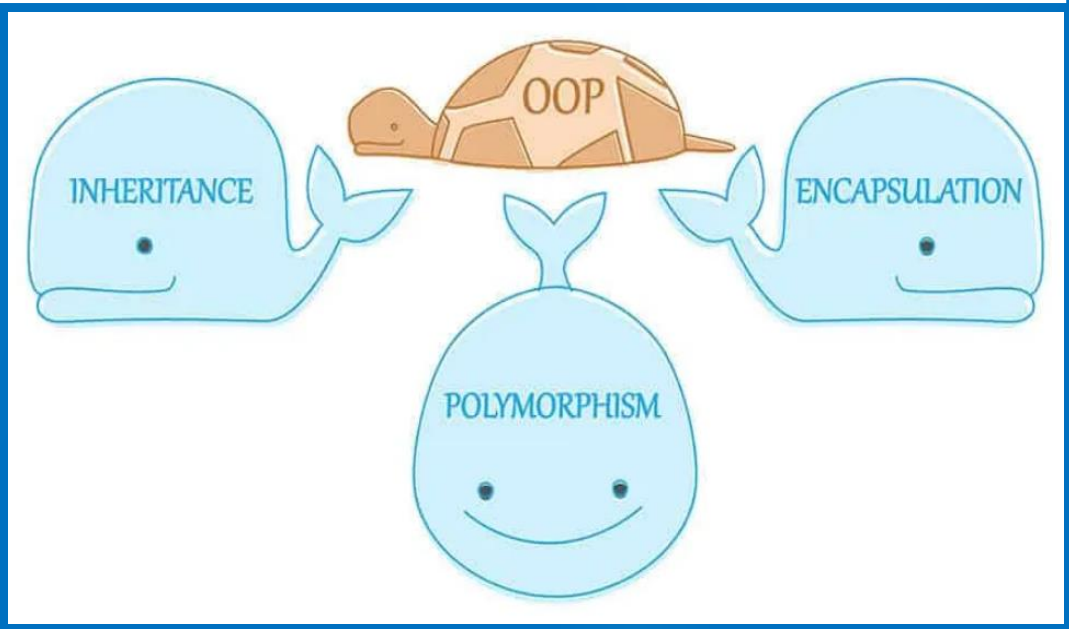
Java is very easy to learn, and its syntax is simple, clean and easy to understand. According to Sun, Java language is a simple programming language because:

- Java *syntax* is based on C++.
- Java has *removed* many complicated and rarely-used features, for example, *explicit pointers*, etc.
- There is no need to remove unreferenced objects because there is an *Automatic Garbage Collection* in Java.

Object-oriented

Java is an *object-oriented* programming language. *Everything in Java is an object*. Object-oriented means we *organize* our *software* as a *combination of* different types of *objects* that incorporates both data and behavior. *Object-oriented programming* (OOPs) is a methodology that simplifies software development and maintenance by providing some rules.

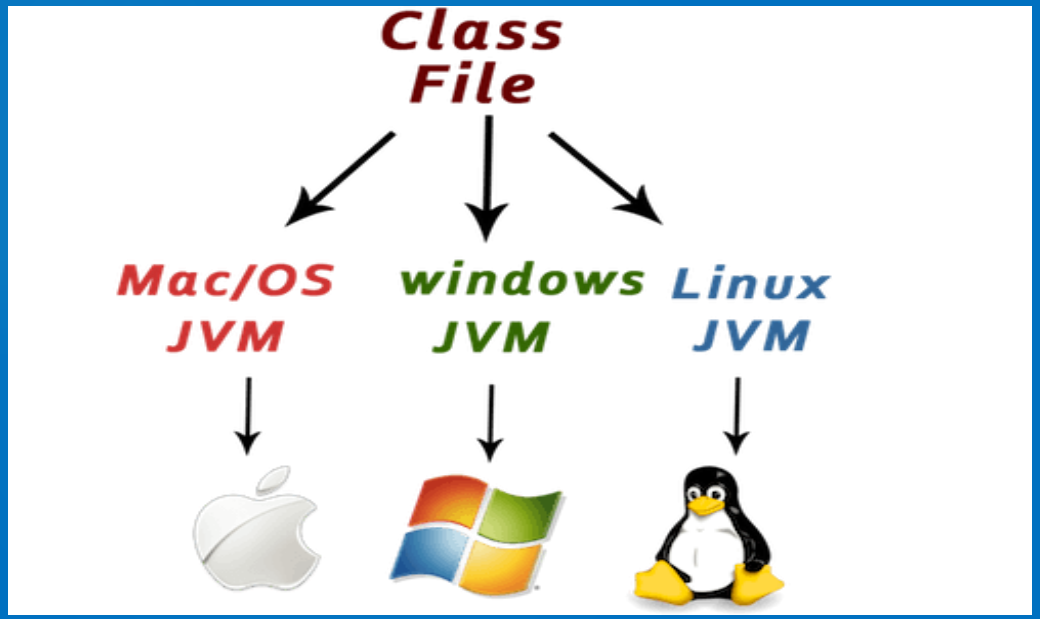


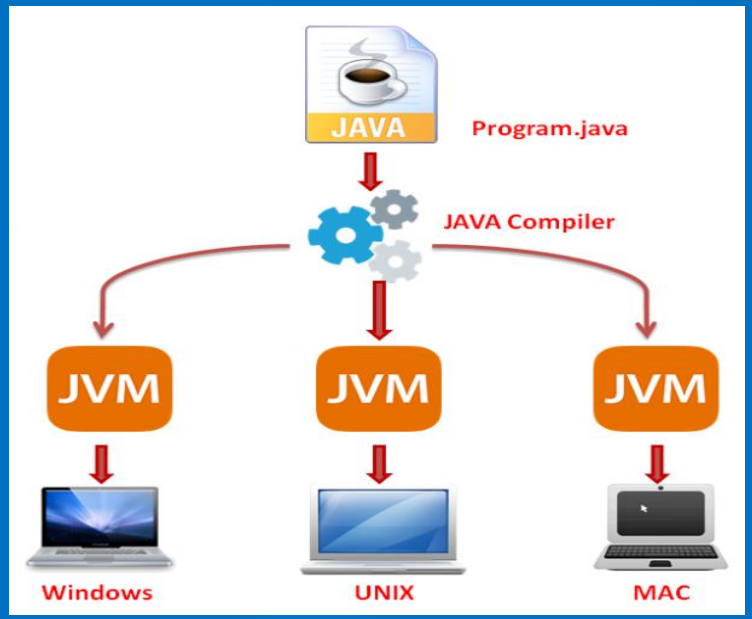


Platform Independent

The Java platform differs from most other platforms in the sense that it is a *software-based platform* that runs on the top of other *hardware-based platforms*. It has two components:

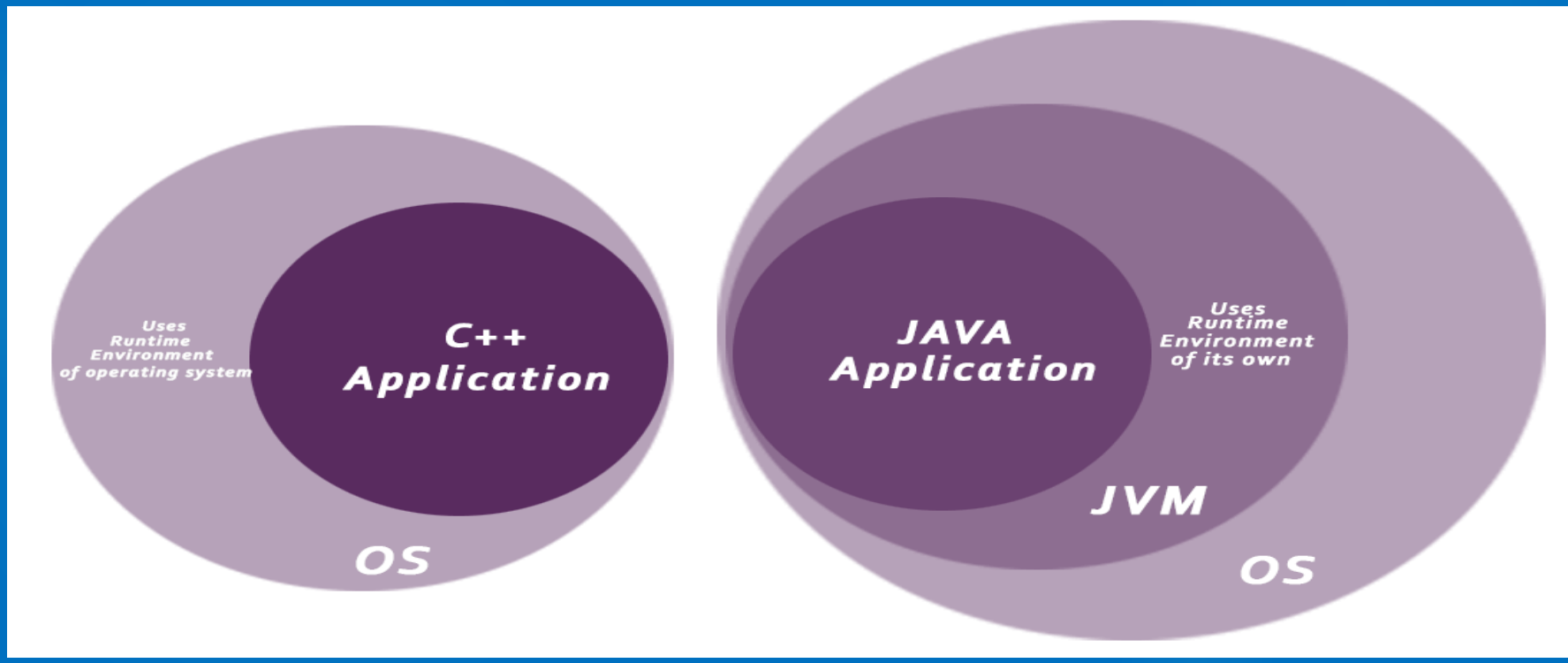
- Runtime Environment
- API(Application Programming Interface)





Secured

Java is best known for its *security*. With Java, we can develop virus-free systems. Java is secured because: (1) No *explicit pointer*; (2) Java Programs run inside a *virtual machine sandbox*; (3) *Bytecode Verifier*; (4) Security Manager; (5) Classloader



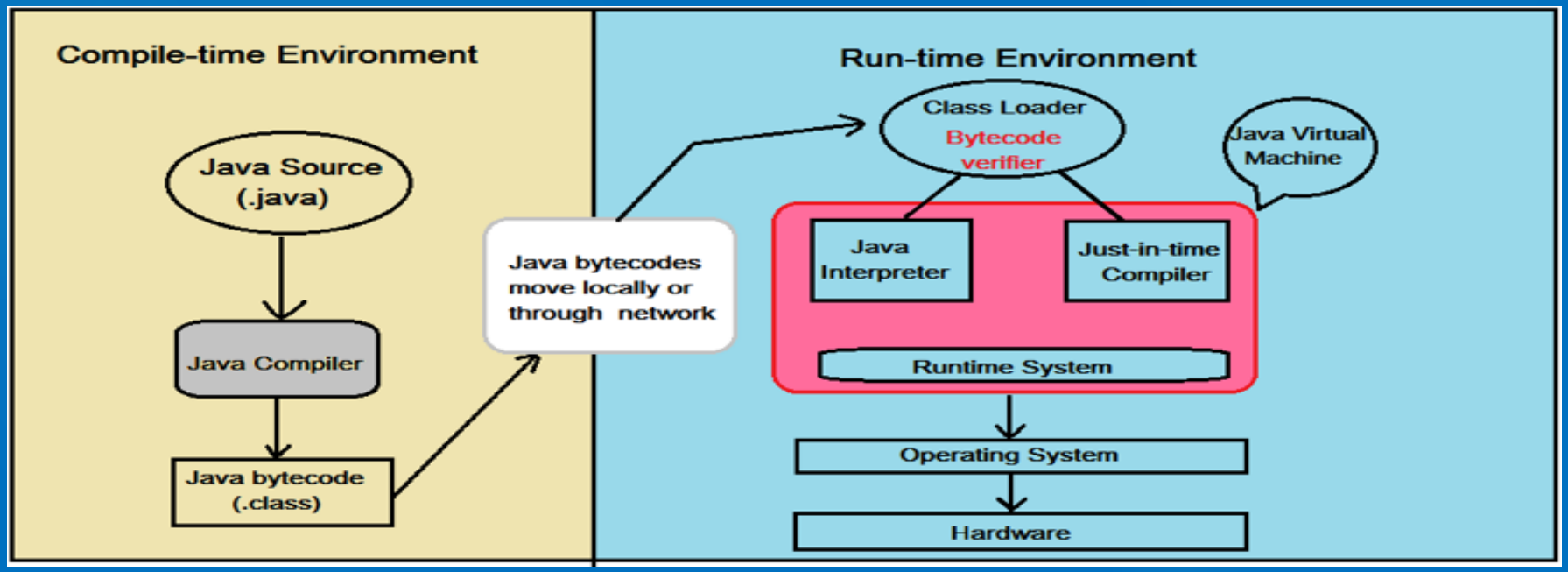
Robust

Robust simply means strong:

- Strong *memory management*.
- Lack of *pointers* that avoids security problems.
- *Automatic garbage collection*.
- *Exception handling* and the *type checking* mechanism.

Interpreted

You need an *interpreter* to run Java programs. The programs are *compiled* into the *Java Virtual Machine code* called *bytecode*. The bytecode is machine-independent and can run on any machine that has a Java interpreter, which is part of the *Java Virtual Machine* (JVM).



Architecture-neutral

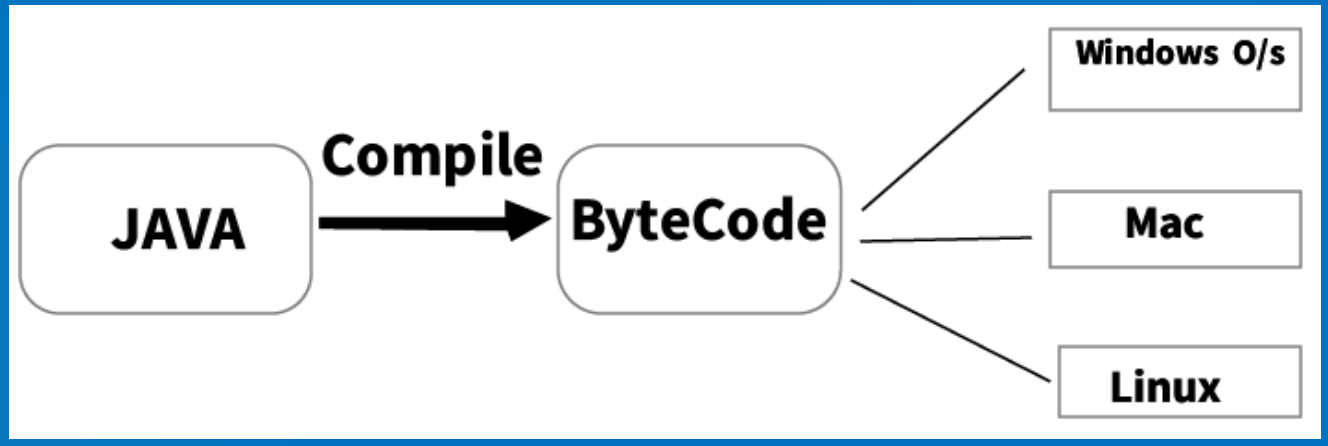
The solution that the Java system adopts to solve the binary-distribution problem is a "*binary code format*" that's independent of hardware architectures, operating system interfaces, and window systems. The format of this system-independent binary code is *architecture neutral*.

In C programming, int data type occupies 2 bytes of memory for *32-bit architecture* and 4 bytes of memory for *64-bit architecture*. However, it occupies 4 bytes of memory for both 32 and 64-bit architectures in Java.

Portable

The primary benefit of the *interpreted byte code* approach is that compiled Java language programs are *portable* to any system on which the Java interpreter and run-time system have been implemented.





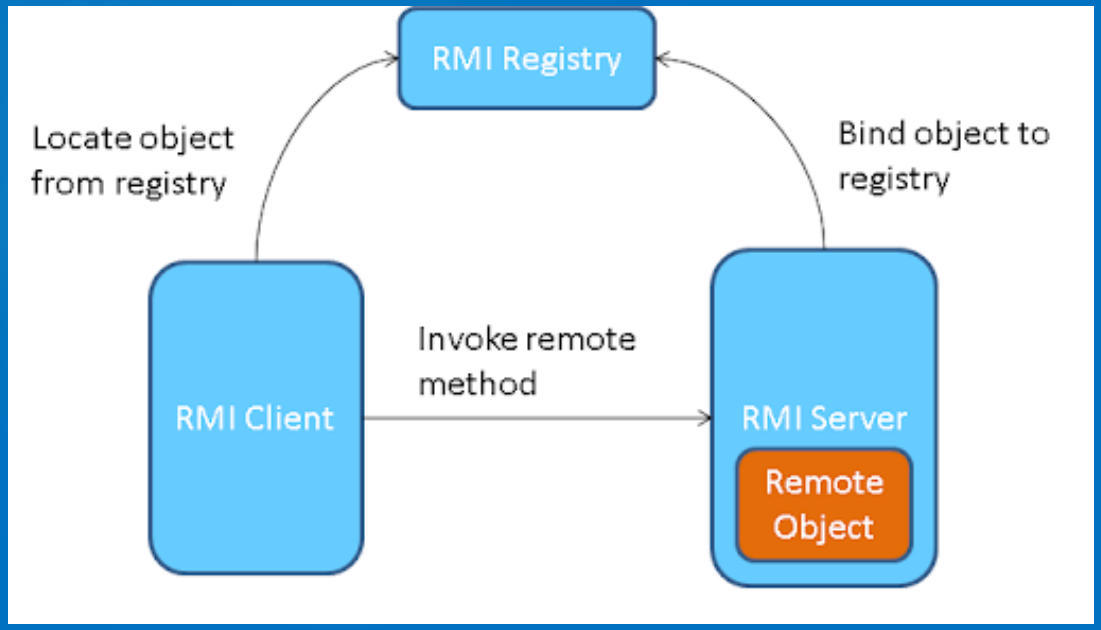
High-performance

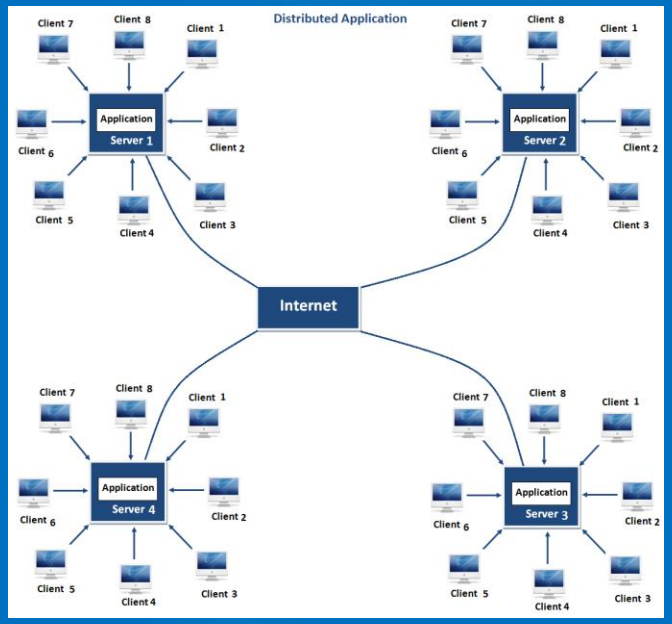
Java is faster than other traditional *interpreted programming languages* because Java bytecode is "*close*" to *native code*. It is still a little bit slower than a *compiled language* (e.g., C++). Java is an interpreted language that is why it is slower than compiled languages, e.g., C, C++, etc.



Distributed

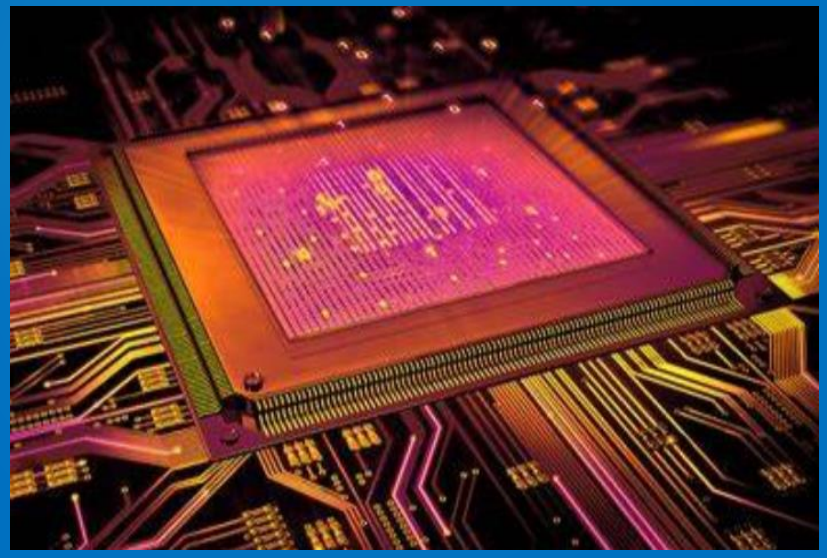
Java is *distributed* because it facilitates users to create *distributed applications* in Java. RMI (Remote Method Invocation) and EJB (Enterprise Java Beans) are used for creating distributed applications. This feature of Java makes us able to access files by *calling the methods from any machine on the internet*.





Multi-threaded

A *thread* is like a *separate program*, executing concurrently. We can write Java programs that *deal with many tasks at once* by defining *multiple threads*. The main advantage of multi-threading is that it doesn't occupy memory for each thread. It shares a common memory area. Threads are important for multi-media, Web applications, etc.





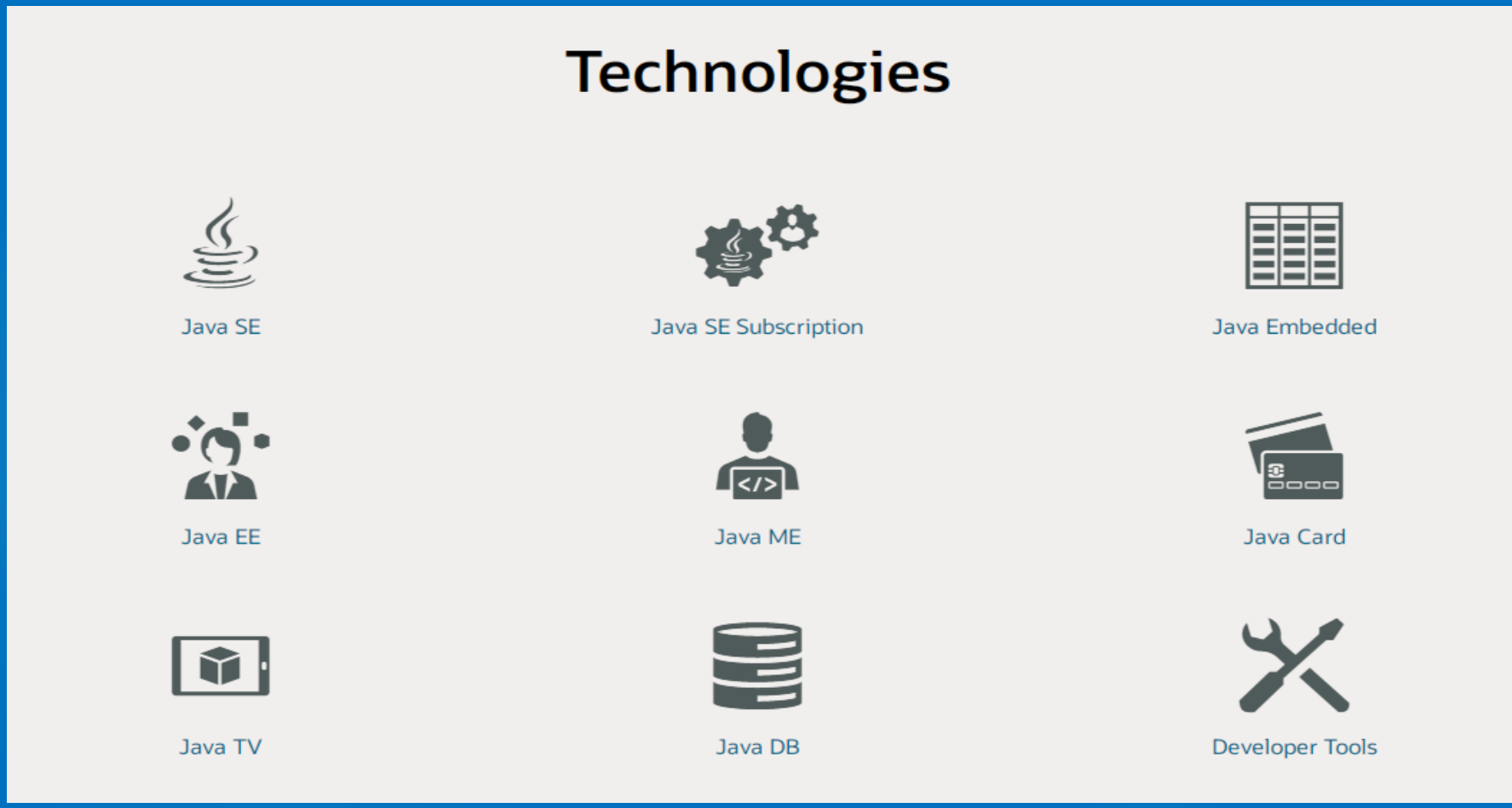
Dynamic

Java is a *dynamic* language. It supports *dynamic loading of classes*. It means classes are loaded on demand. It also supports functions from its native languages, i.e., C and C++.

Java supports *dynamic compilation* and automatic memory management (garbage collection).

Java Platform

https://www.oracle.com/java/technologies/



Download JDK

https://www.oracle.com/java/technologies/javase-downloads.html

Java SE 8u261

Java SE 8u261 includes important bug fixes. Oracle strongly recommends that all Java SE 8 users upgrade to this release.

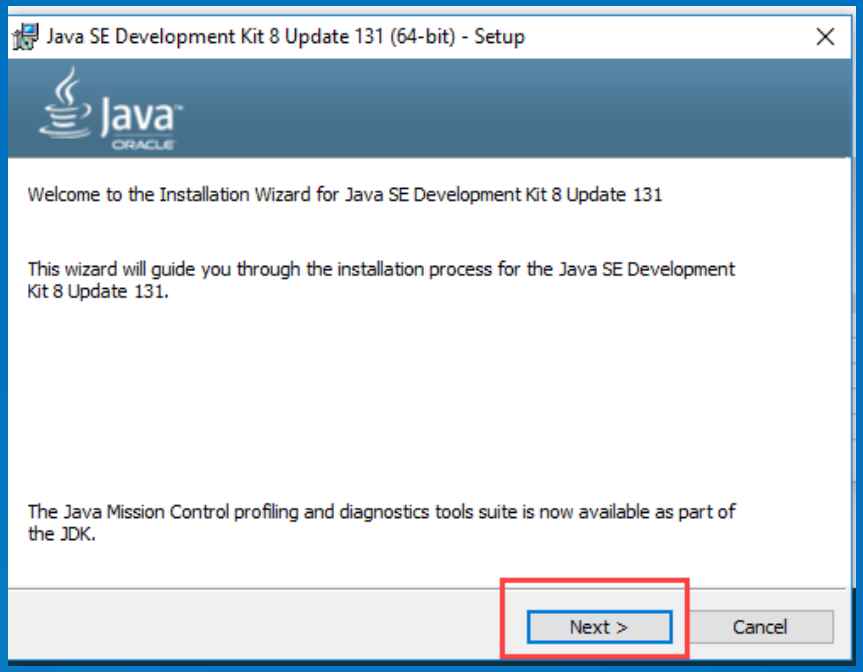
- Documentation
- Installation Instructions
- Release Notes
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- Java SE Licensing Information User Manual
  - Includes Third Party Licenses
- Certified System Configurations
- Readme Files
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  - JRE ReadMe

Oracle JDK

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Windows x86	154.52 MB	jdk-8u261-windows-i586.exe
Windows x64	166.28 MB	jdk-8u261-windows-x64.exe

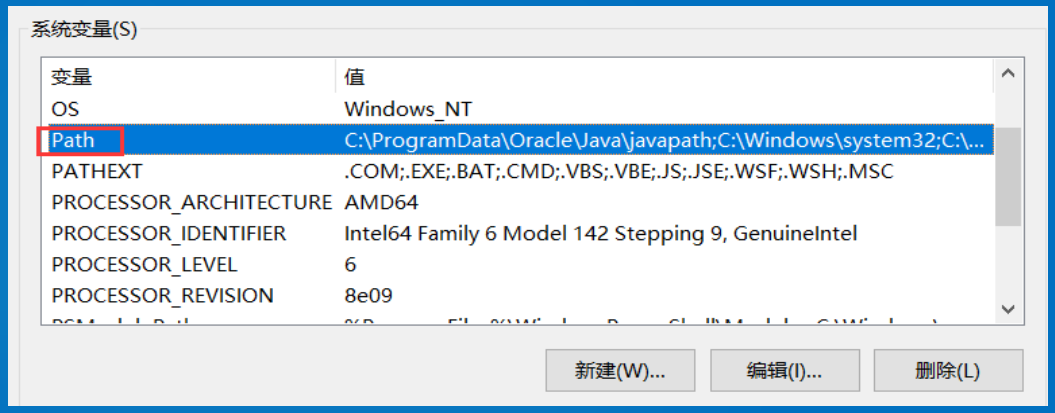
JDK Installation

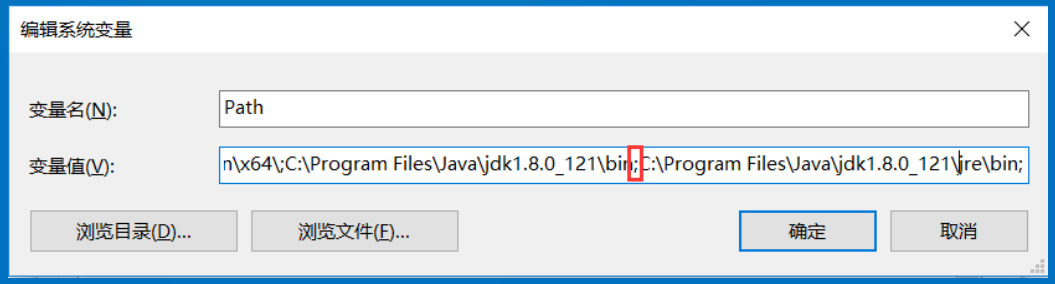


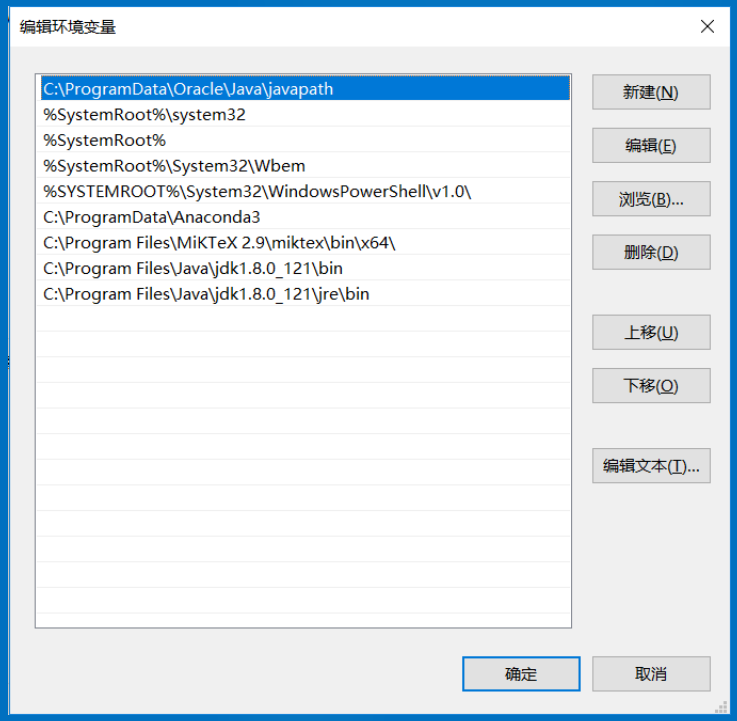


JDK Configuration

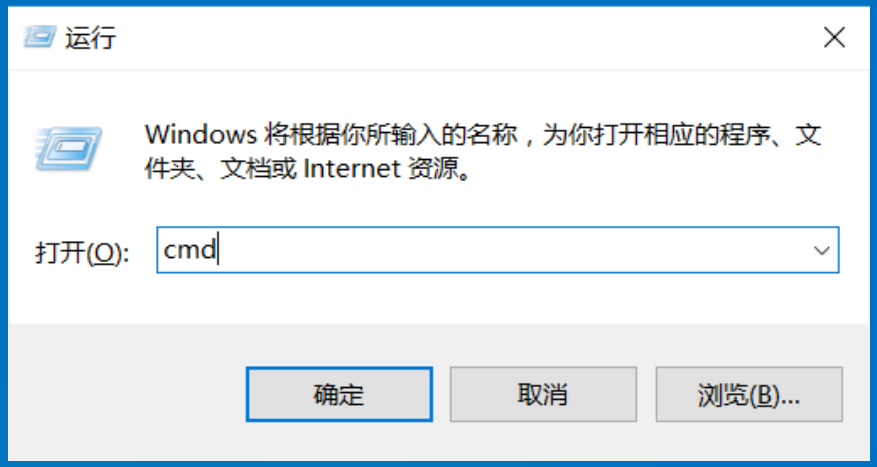
An ***environment variable*** is a dynamic-named value that can affect the way running processes will behave on a computer.
PATH is an *environment variable* on *Unix-like operating systems*, *DOS*, *OS/2*, and *Microsoft Windows*, specifying a set of *directories* where executable programs are located.

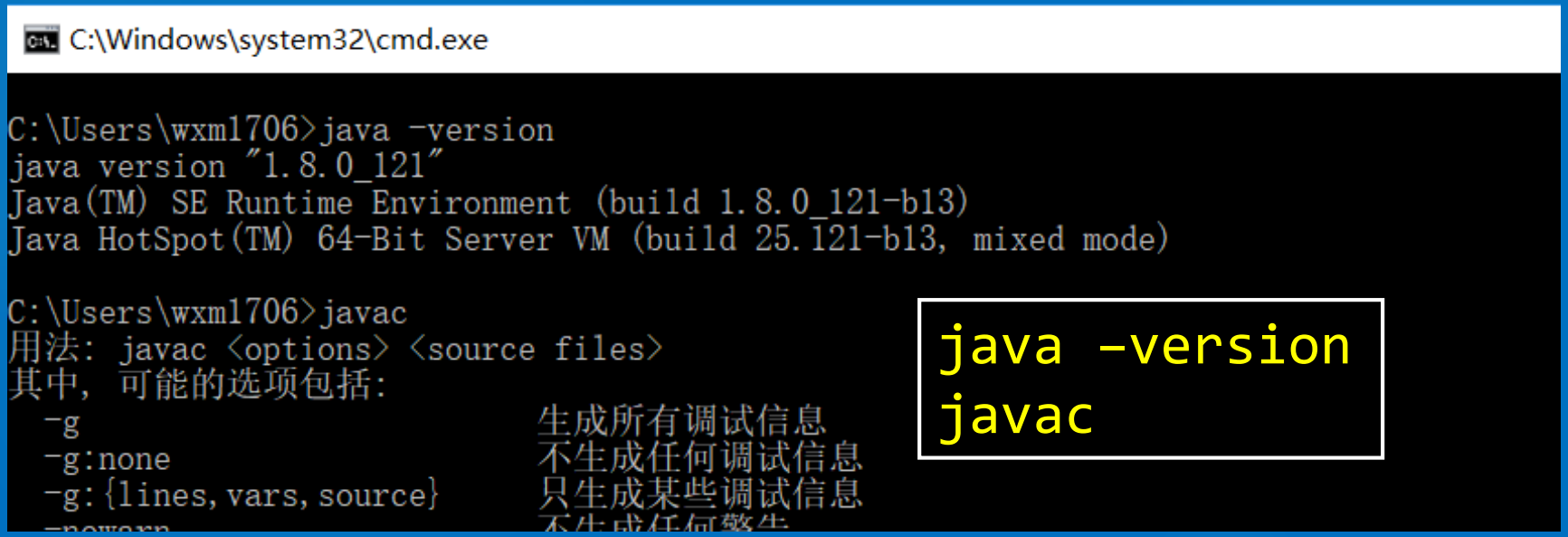






Environment Test



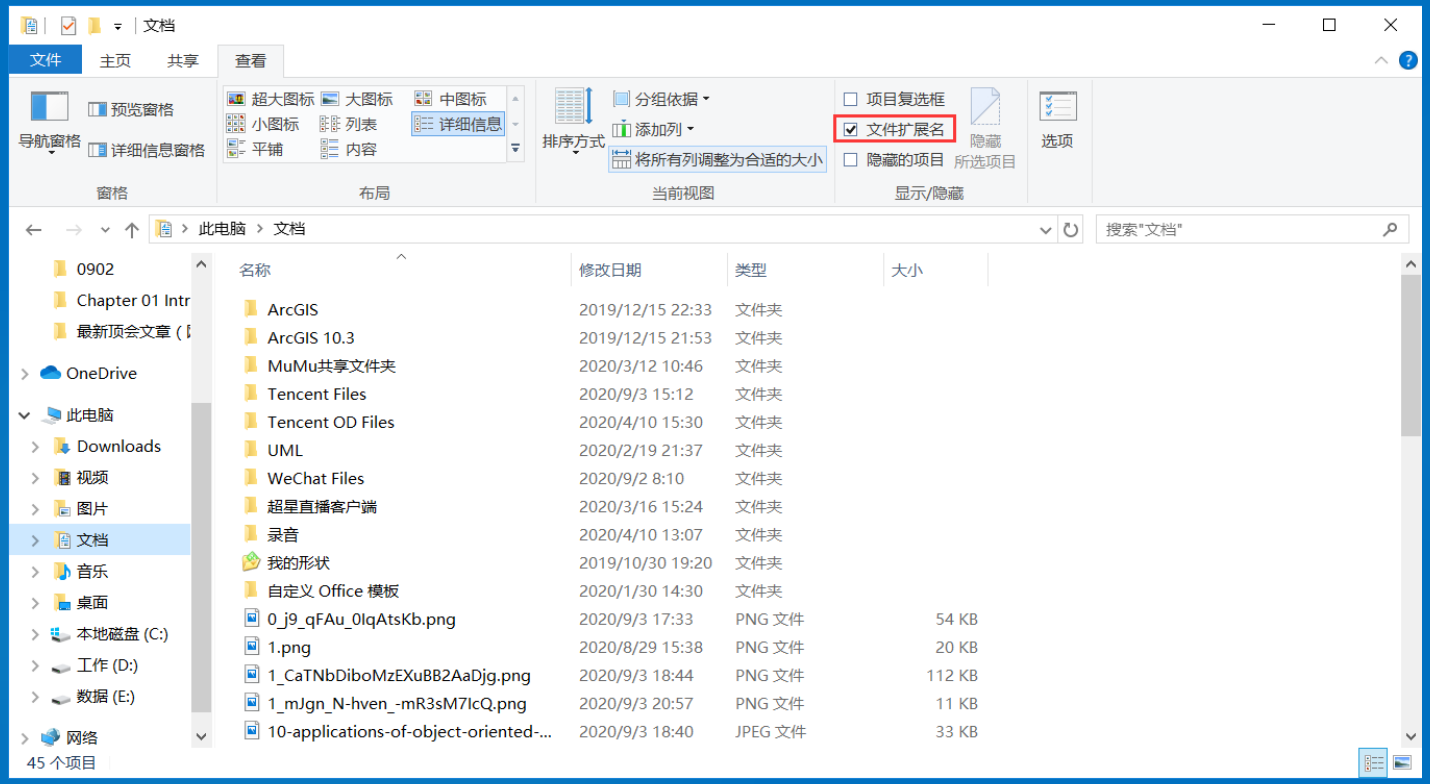


Try to install and configure JDK

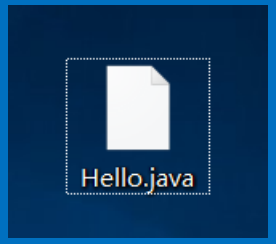


A Simple Java Program

Show file extension:



Create a txt file named Hello.java



A Simple Java Program

Edit codes in the file:

```
class Hello{
    public static void main(String[] args){
        if(args.length > 0){
            System.out.println("Hi "+args[0]+"!");
        }else{
            System.out.println("Hi!");
        }
    }
}
```

A Simple Java Program

Compile and run the program:

```
C:\Users\wxm1706>cd desktop

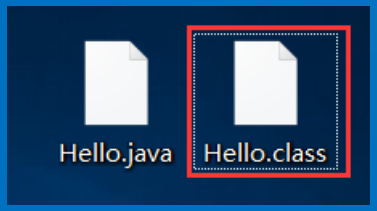
C:\Users\wxm1706\Desktop>
C:\Users\wxm1706\Desktop>
C:\Users\wxm1706\Desktop>javac Hello.java

C:\Users\wxm1706\Desktop>java Hello xiaoming
Hi xiaoming!

C:\Users\wxm1706\Desktop>
```

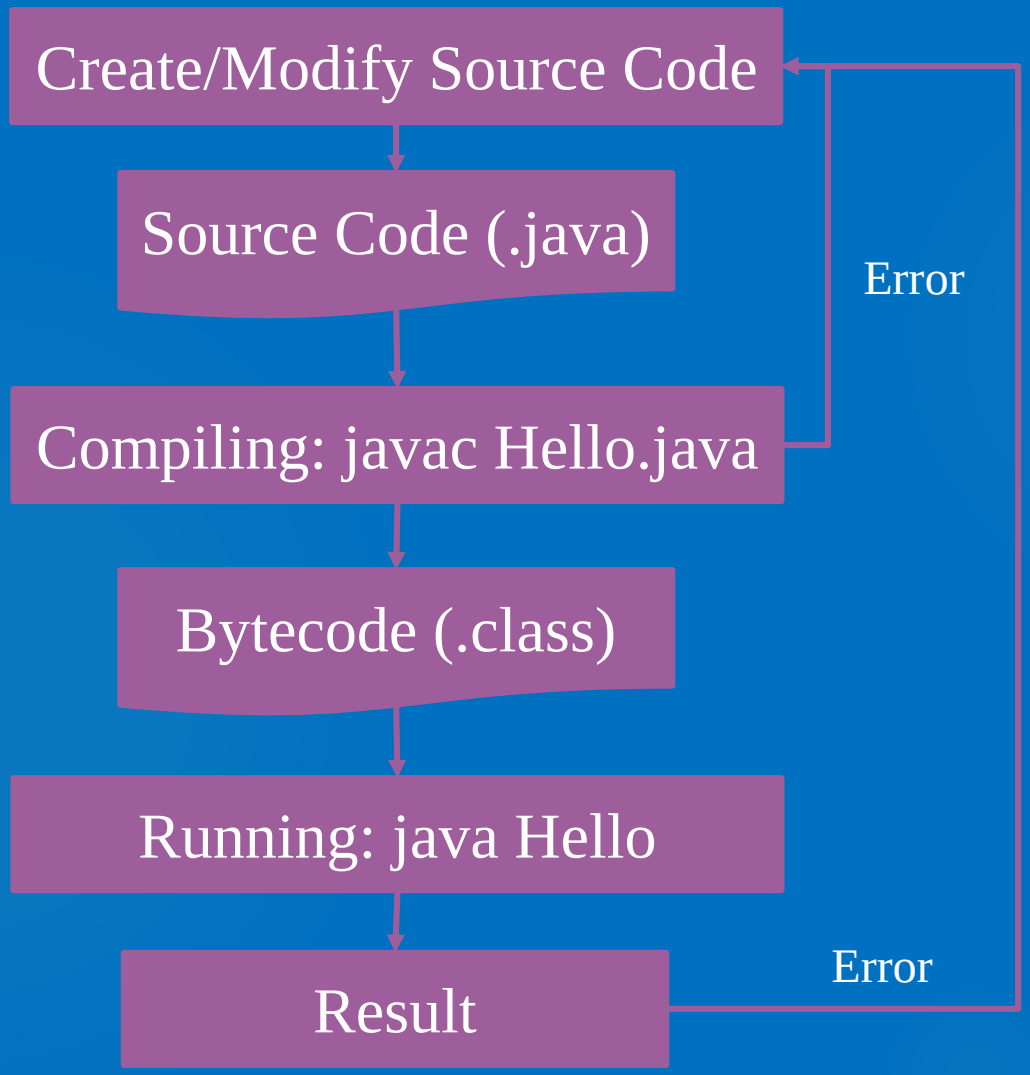
Switch to the directory where the source code is located

Compile the source file using the javac command



Run the program using the java command

Programming Process



Comments

A *line comment* is preceded by two slashes (//) in a line. A *paragraph comment* is enclosed between /* and */ in one or multiple lines.

```
/*
 * Version: 1.0
 * Author: xiaoming
 *
*/
class Hello{
    public static void main(String[] args){
        System.out.println("Hello"); //print something
    }
}
```

Comments

javadoc comment: javadoc comments begin with /** and end with */. They are used for documenting classes, data, and methods. They can be extracted into an HTML file using JDK's javadoc command.

String.class *Client.java

```
39 import java.util.regex.Pattern;
40 import java.util.regex.PatternSyntaxException;
41
42 /**
43  * The {@code String} class represents character strings. All
44  * string literals in Java programs, such as {@code "abc"}, are
45  * implemented as instances of this class.
46  * <p>
47  * Strings are constant; their values cannot be changed after they
48  * are created. String buffers support mutable strings.
49  * Because String objects are immutable they can be shared. For example:
50  * <blockquote><pre>
```

compact1, compact2, compact3

java.lang

Class String

java.lang.Object
    java.lang.String

All Implemented Interfaces:

Serializable, CharSequence, Comparable<String>

public final class String
extends Object
implements Serializable, Comparable<String>, CharSequence

The String class represents character strings. All string literals in Java programs, such as "abc",

javadoc

Reserved Words

Reserved words or keywords are words that have a *specific meaning* to the *compiler* and *cannot be used* for other purposes in the program. For example, when the compiler sees the word *class*, it understands that the word after class is the name for the class.
https://docs.oracle.com/javase/specs/jls/se11/html/jls-3.html#jls-3.9

abstract	continue	for	new	switch
assert	default	if	package	synchronized
boolean	do	goto	private	this
break	double	implements	protected	throw
byte	else	import	public	throws
case	enum	instanceof	return	transient
catch	extends	int	short	try
char	final	interface	static	void
class	finally	long	strictfp	volatile
const	float	native	super	while
_ *(underscore)*				

true and false are not keywords, but rather boolean literals; null is not a keyword, but rather the *null literal*

Modifiers

Java uses certain *reserved words* called modifiers that *specify the properties* of the *data*, *methods*, and *classes* and *how* they can be used. Examples of modifiers are *public* and *static*. Other modifiers are *private*, *final*, *abstract*, and *protected*. A public datum, method, or class can be accessed by other programs. A private datum or method cannot be accessed by other programs.

Modifiers: *public, protected, private*
Class Modifiers: *abstract, final, strictfp*
Field Modifiers: *static, final, transient, volatile*
Method Modifiers: *abstract, static, final, native, strictfp, synchronized*
Interface Modifiers : *abstract, strictfp*

Statements

A *statement* represents an *action* or a sequence of actions. The statement System.out.println(“Hi!”) is a statement to display the greeting “Hi!”. Every statement in Java ends with a semicolon (;).

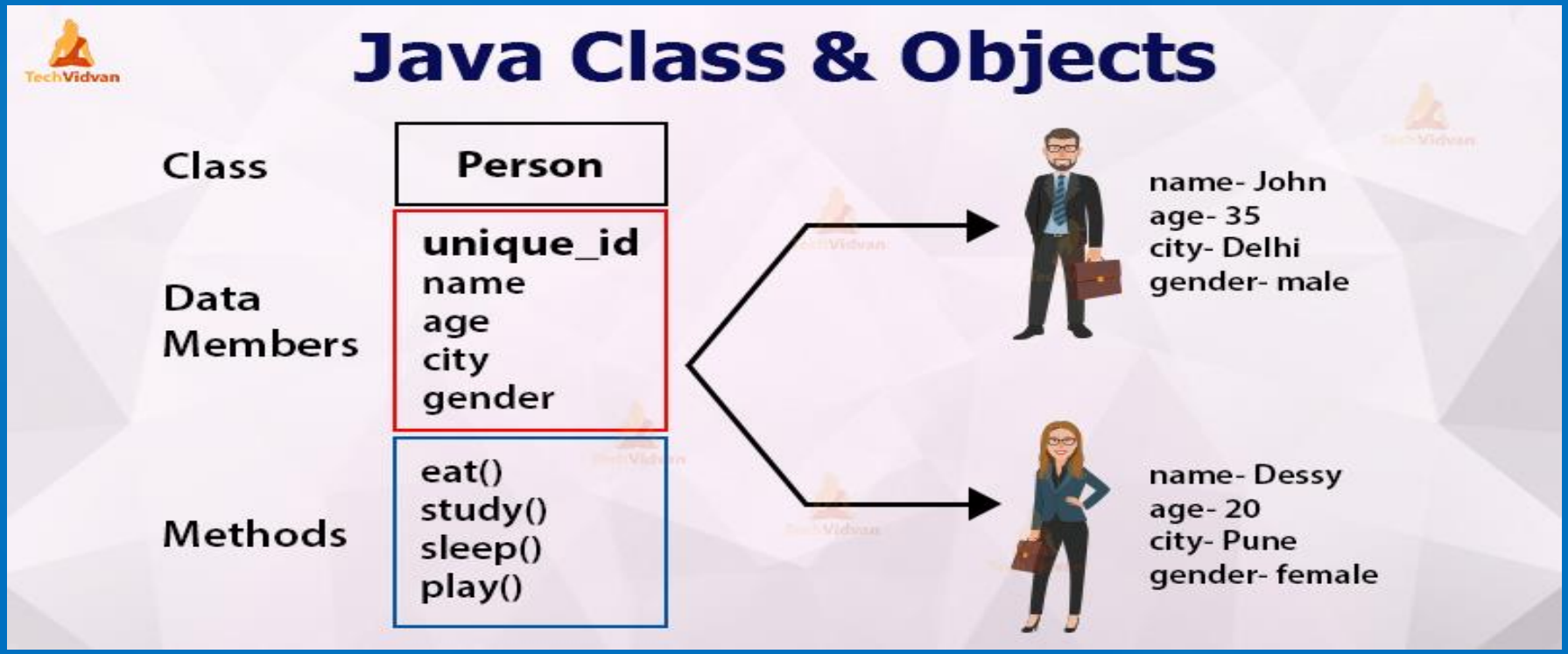
Blocks

A *block* is a sequence of statements, local class declarations, and local variable declaration statements within *braces*. It groups components of a program.

```
class Hello{
    {
        System.out.println("something");
    }
    public static void main(String[] args){
        System.out.println("Hello");
        {
            System.out.println("something");
        }
    }
}
```

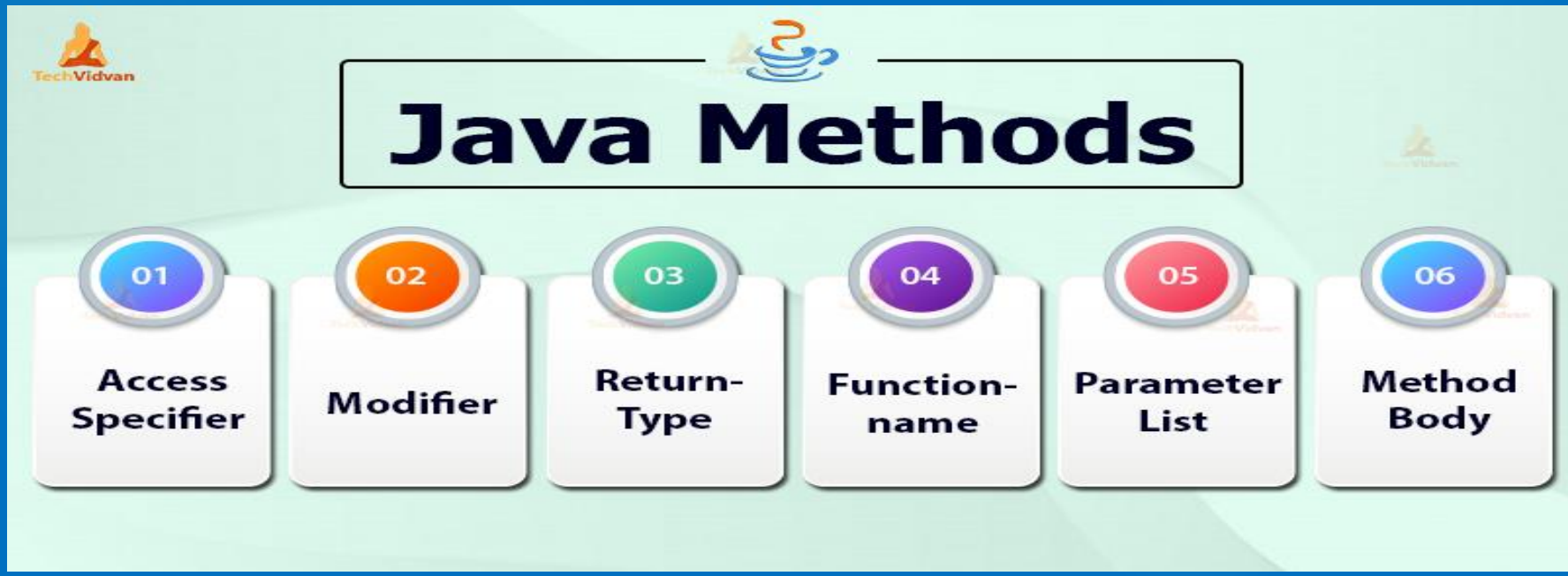
Classes

The class is the essential Java construct. A class is a *template* or *blueprint* for *objects*. To program in Java, you must understand classes and be able to write and use them.



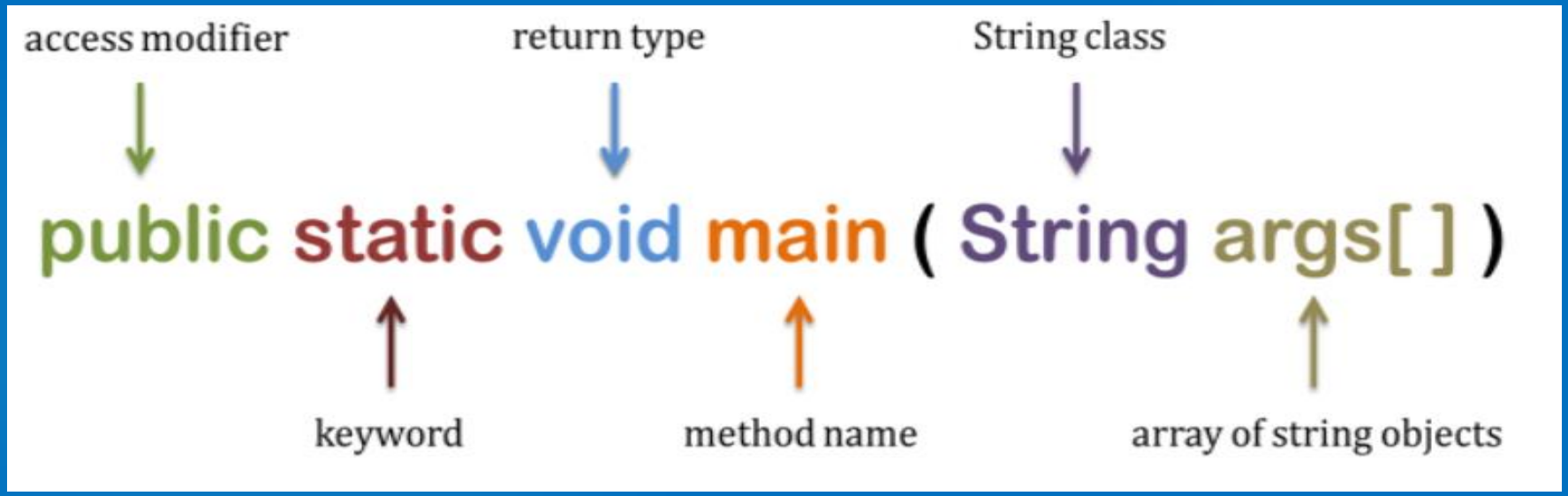
Methods

What is System.out.println? It is a *method*: a collection of statements that performs a sequence of operations to display a message on the console. It can be used even without fully understanding the details of how it works. It is used by invoking a statement with a string argument. The string argument is enclosed within parentheses.



main()

The main method provides the control of program flow. The Java interpreter executes the application by invoking the main method.



Naming Conventions

Class, Interface: First letter uppercase

class HelloWorld

Argument, Method, Variable: First letter lowercase

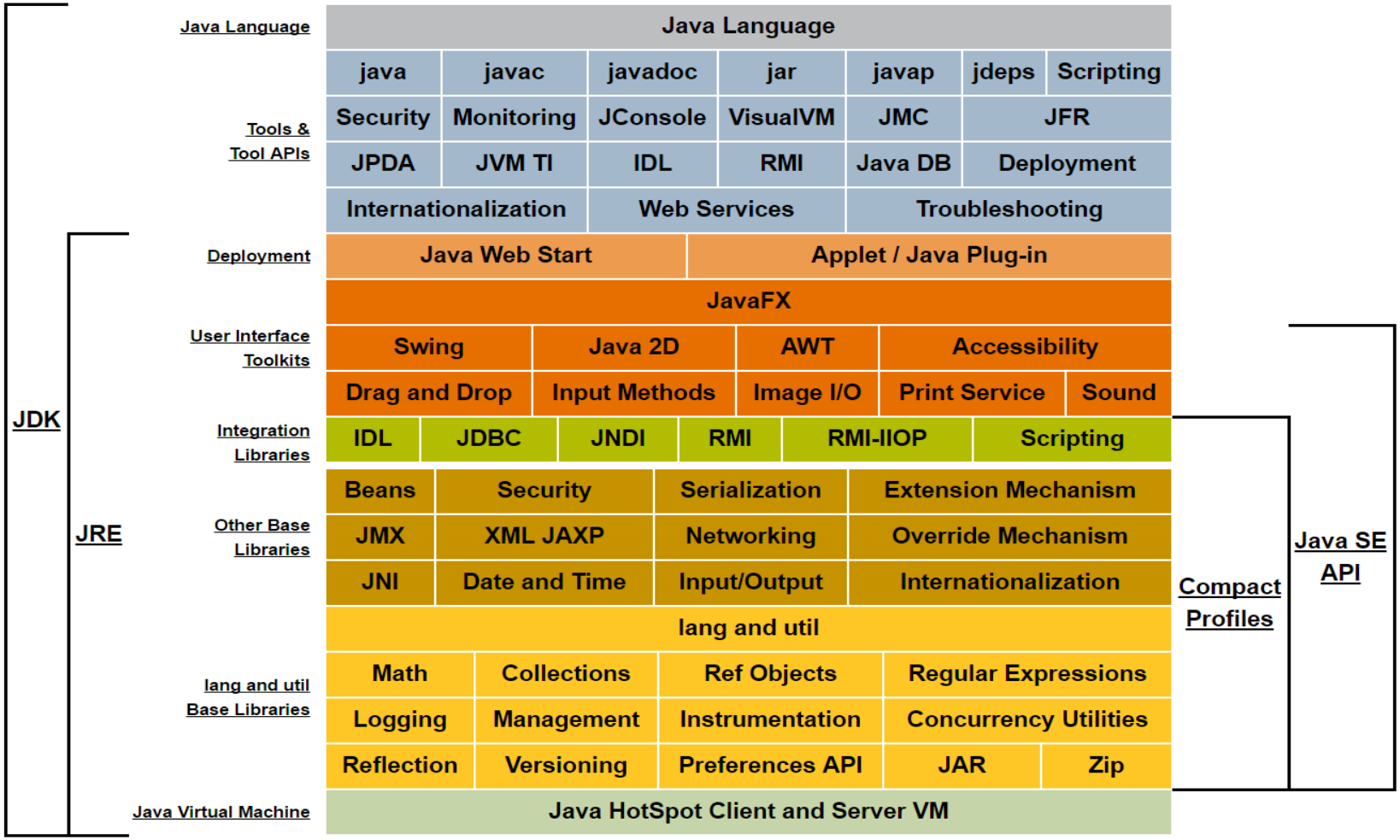
int myNumber;

public void addTwoNumber(int a, int b)

https://www.oracle.com/technetwork/java/codeconventions-150003.pdf

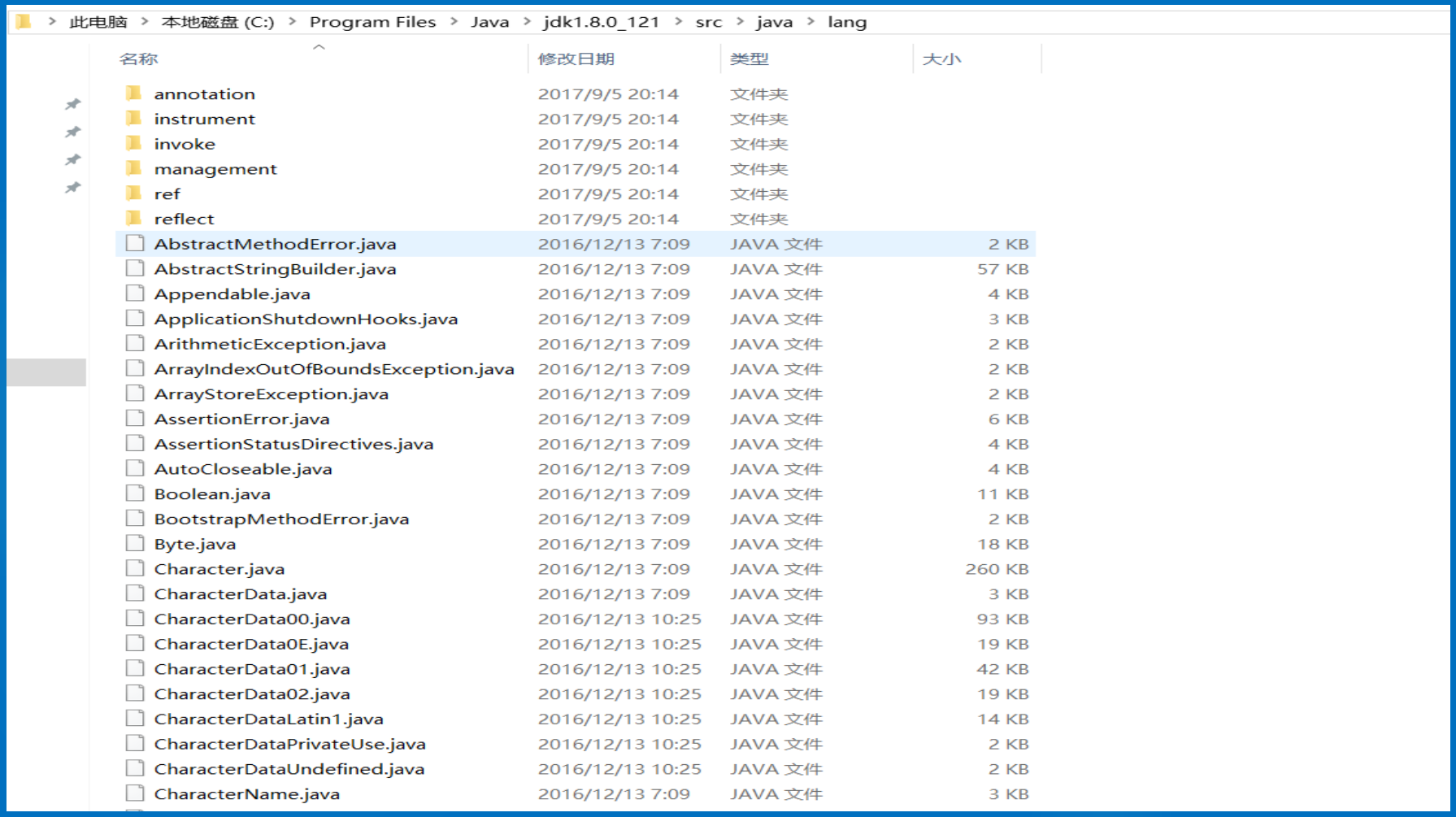
Study JDK Source Code

http://docs.oracle.com/javase/8/docs/index.html



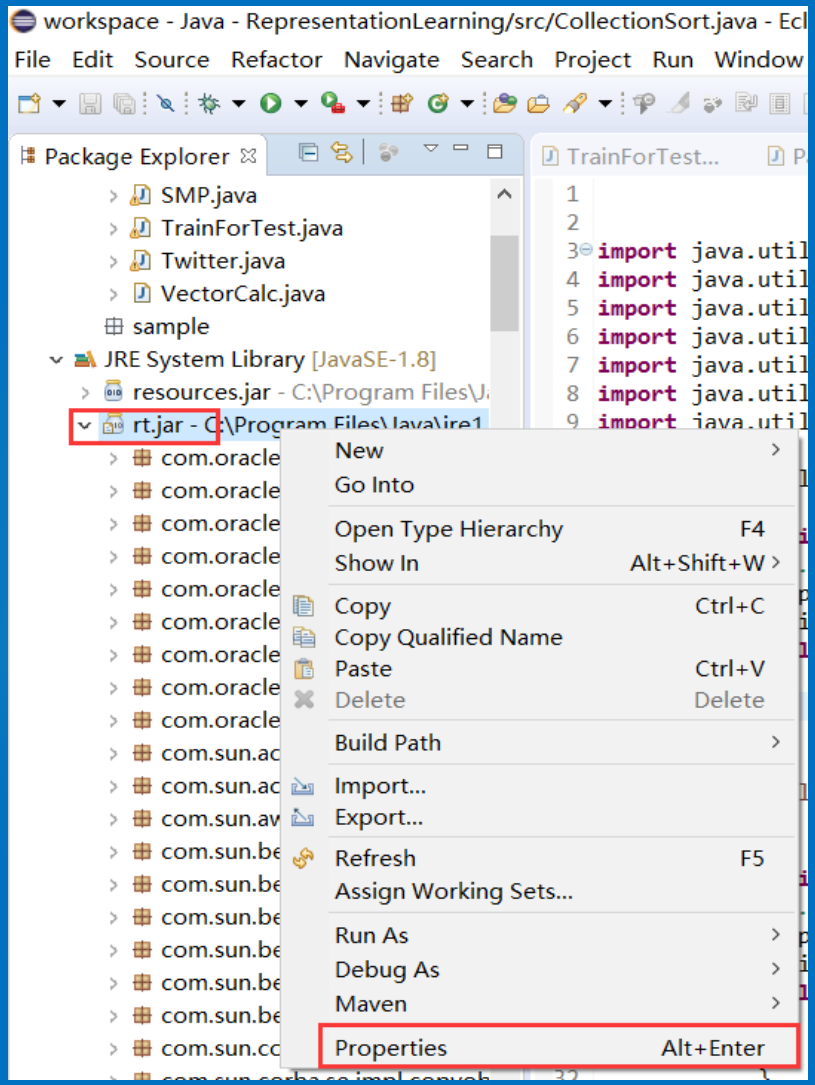
Study JDK Source Code

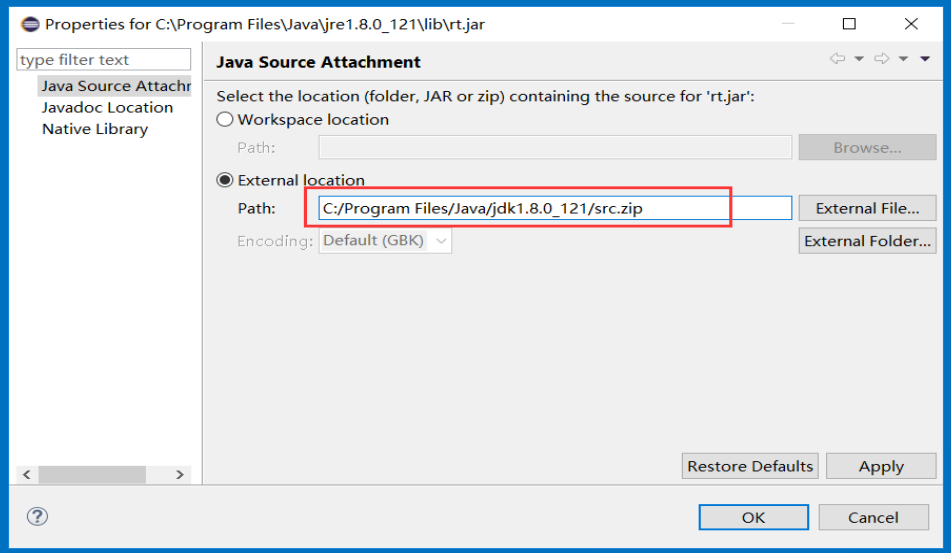
C:\Program Files\Java\jdk\src.zip

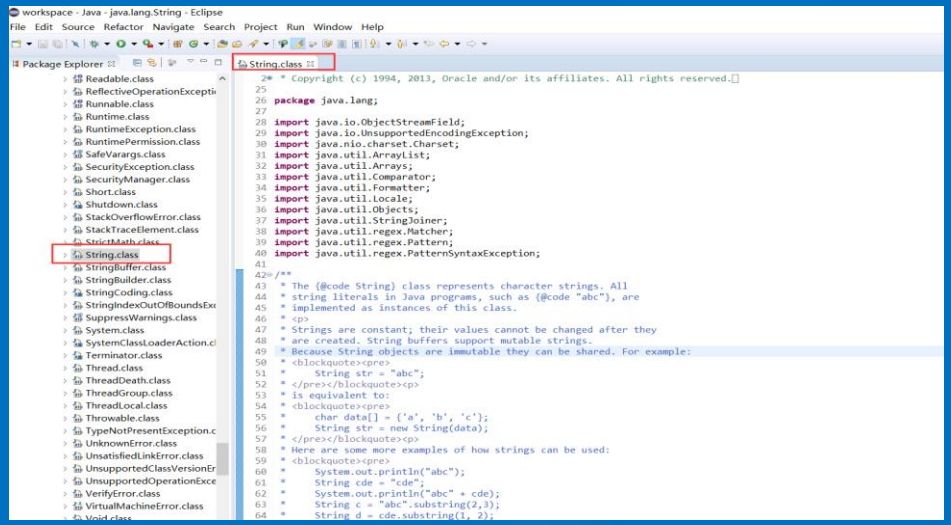


Study JDK Source Code

How to view source code in Eclipse?







Code Editor Vs IDE

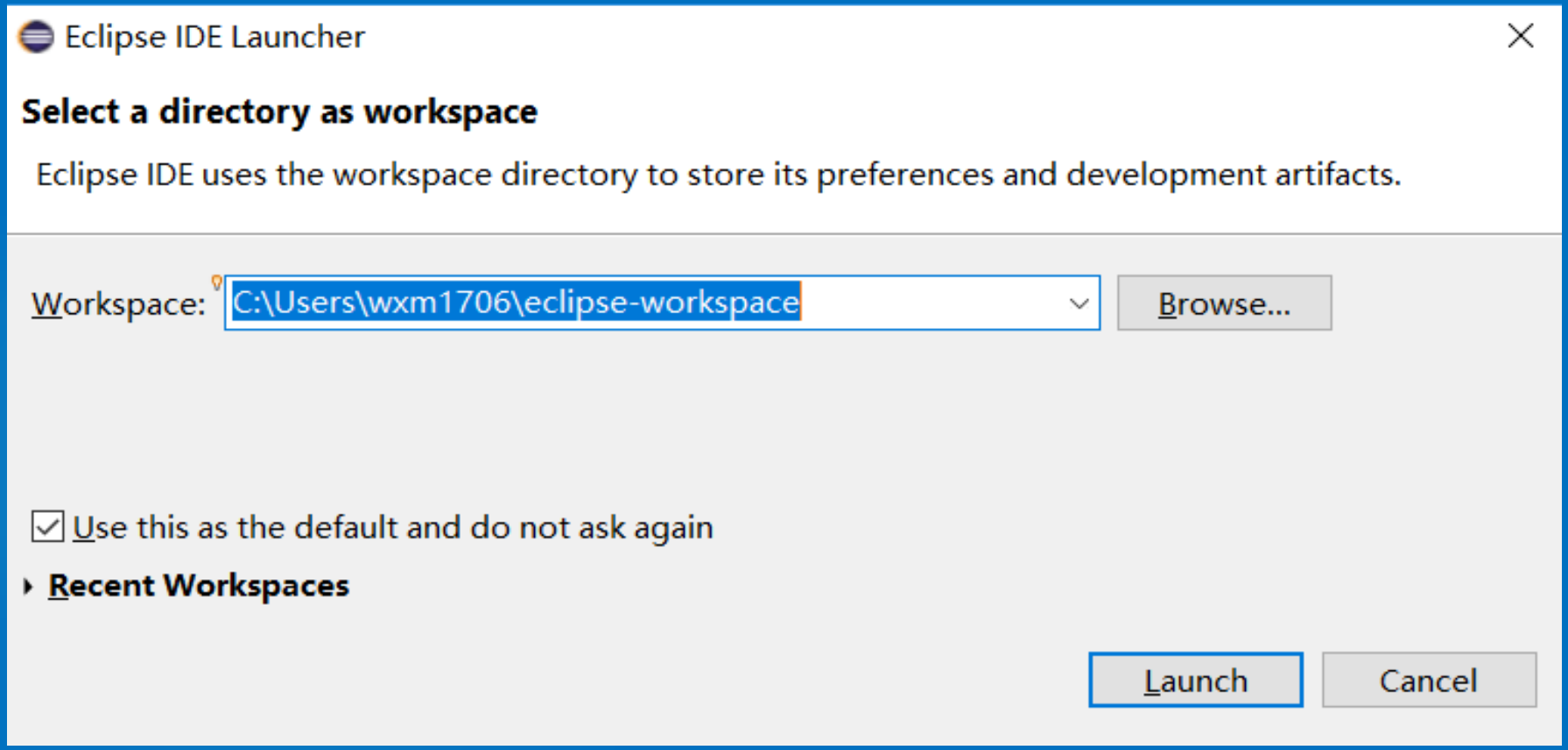


Develop with a text Editor

- Make sure you are editing the *correct file*
- Switch to *the directory where the source file is located* before executing the compilation command
- Modify the source code and *save* it before *recompiling*

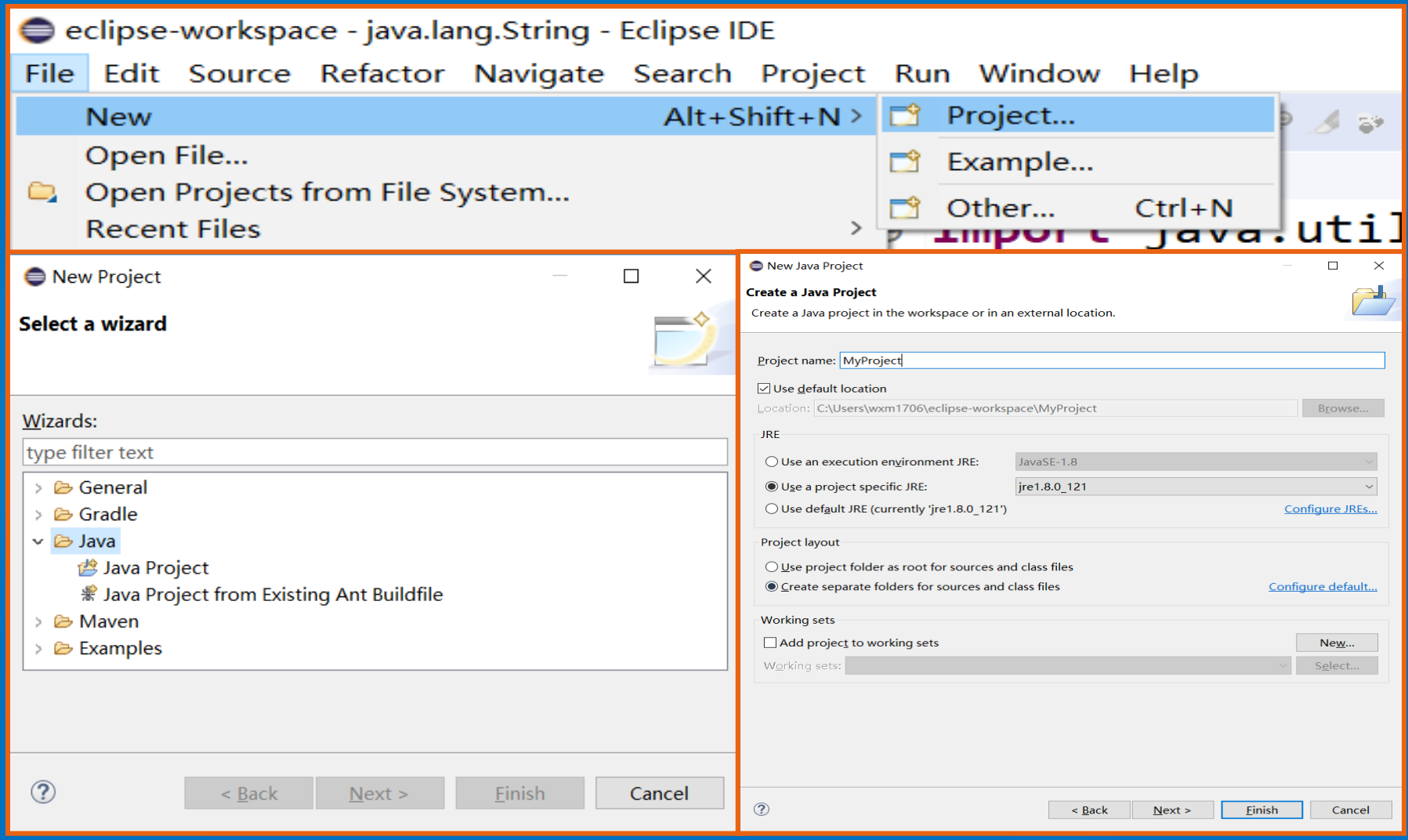
Develop with Eclipse

Workspace



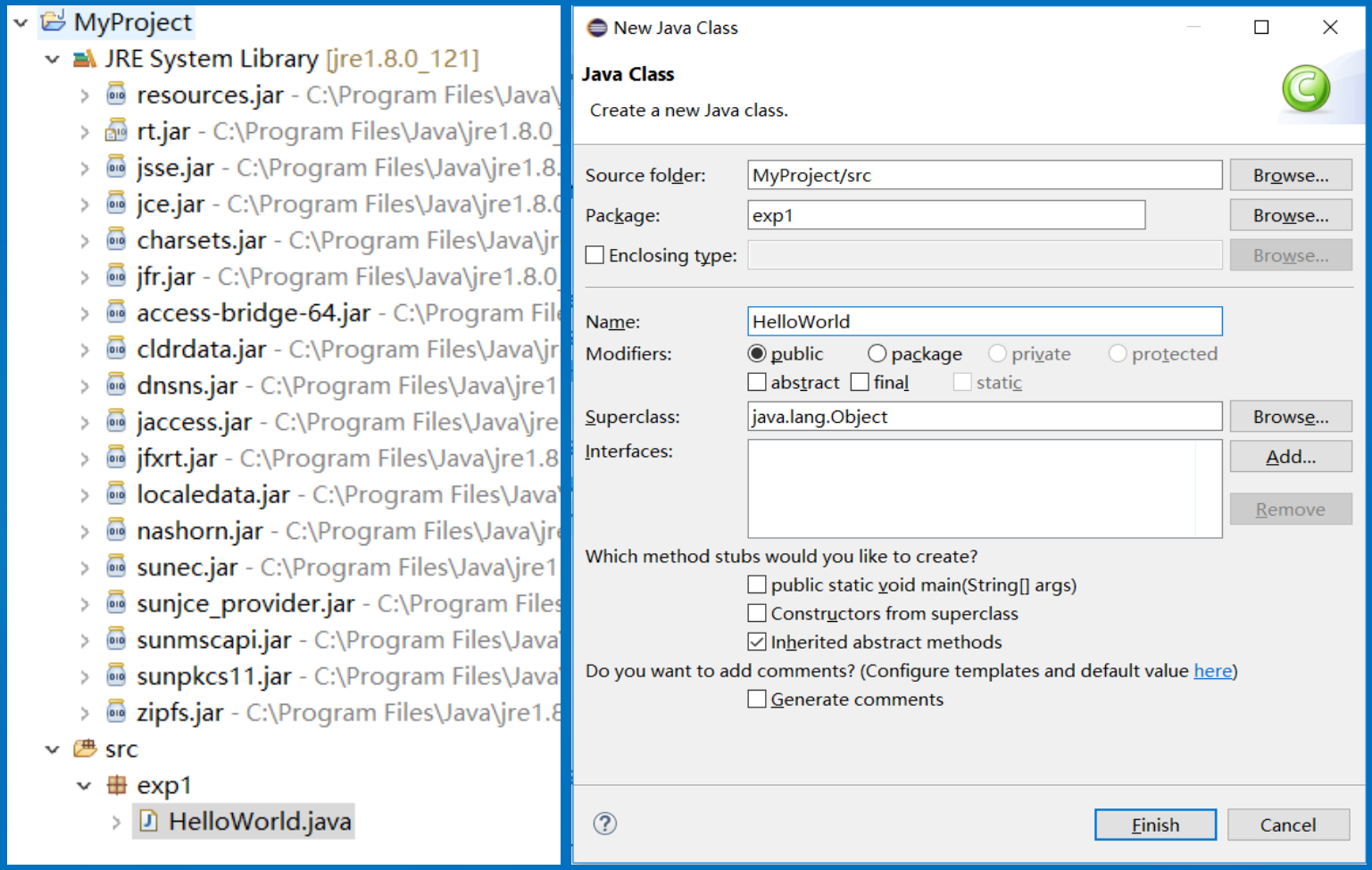
Develop with Eclipse

Java Project



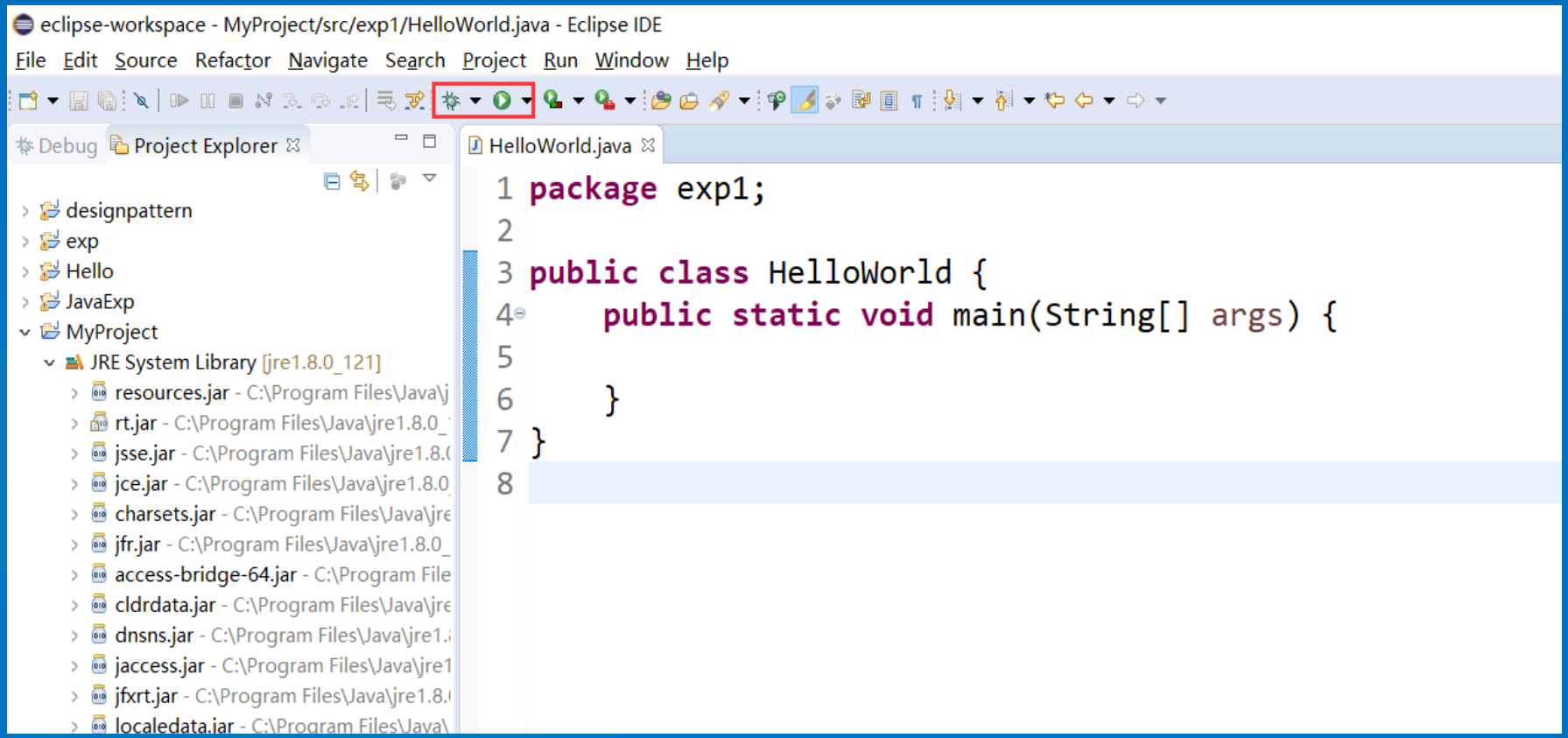
Develop with Eclipse

Create a new class



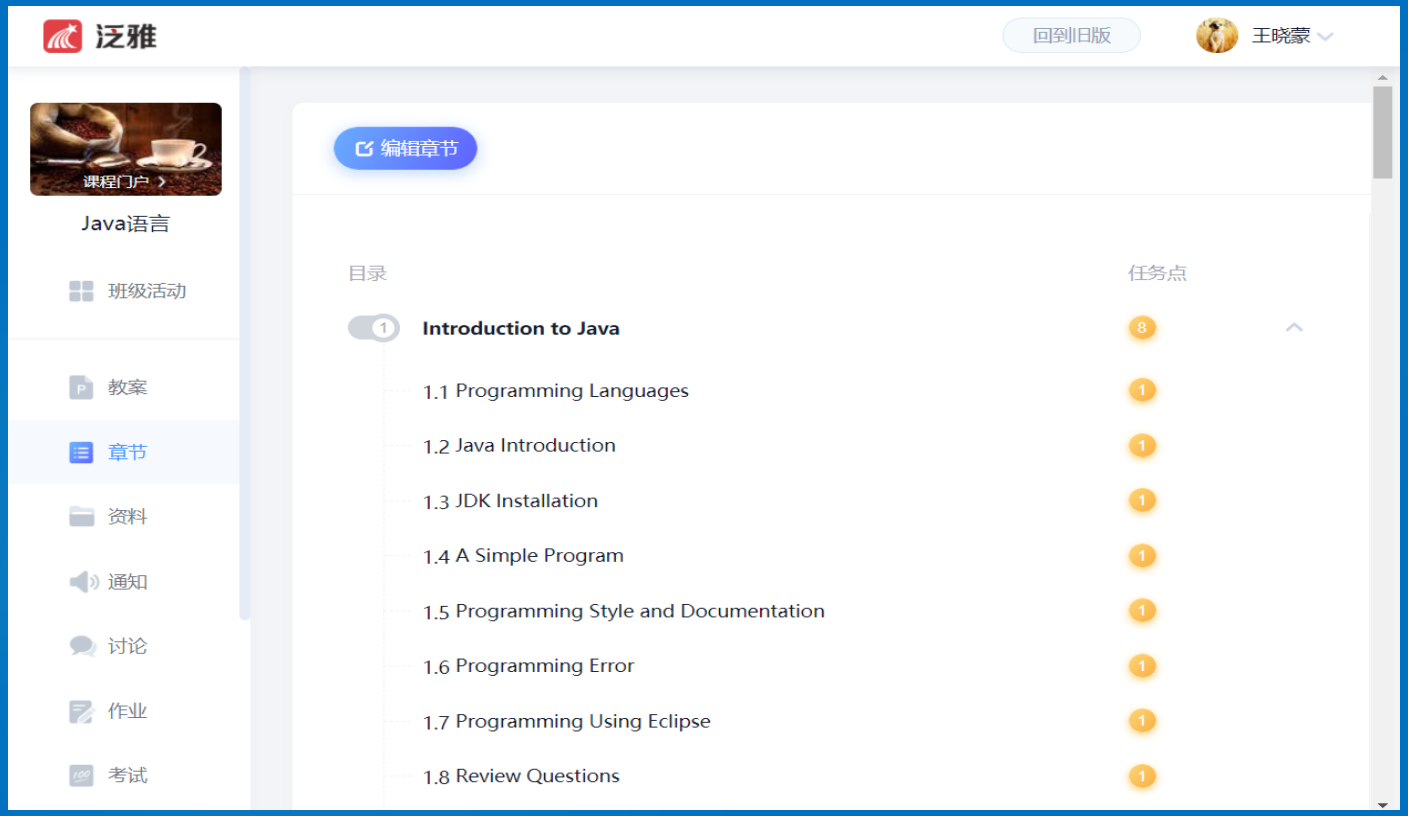
Develop with Eclipse

Coding and Compiling



E-Learning

https://mooc1.chaoxing.com/course/200212070.html

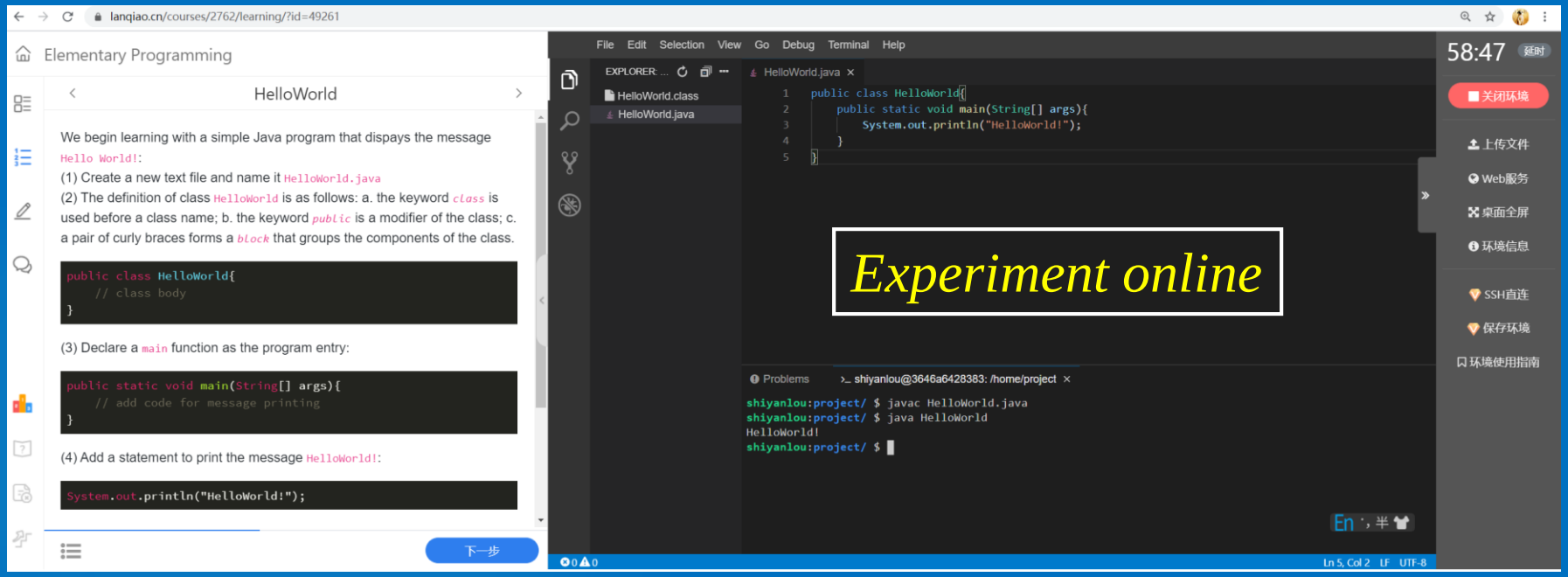




E-Learning

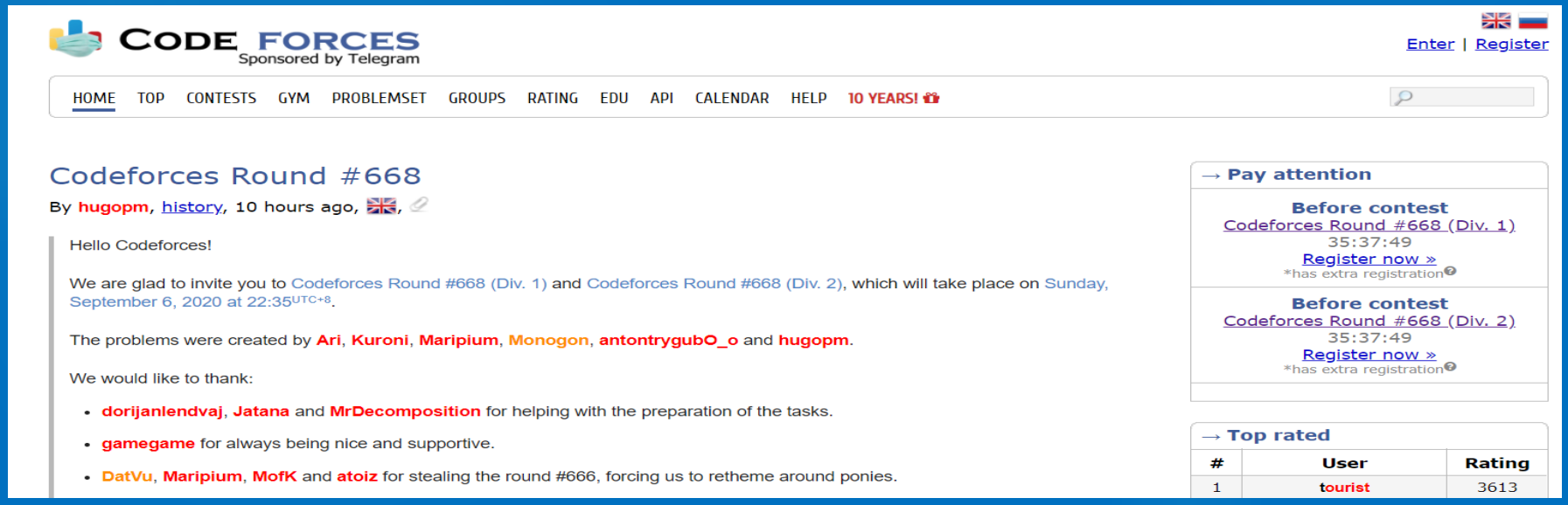
https://www.lanqiao.cn/courses/2762

邀请码：ZF0XA4Y1

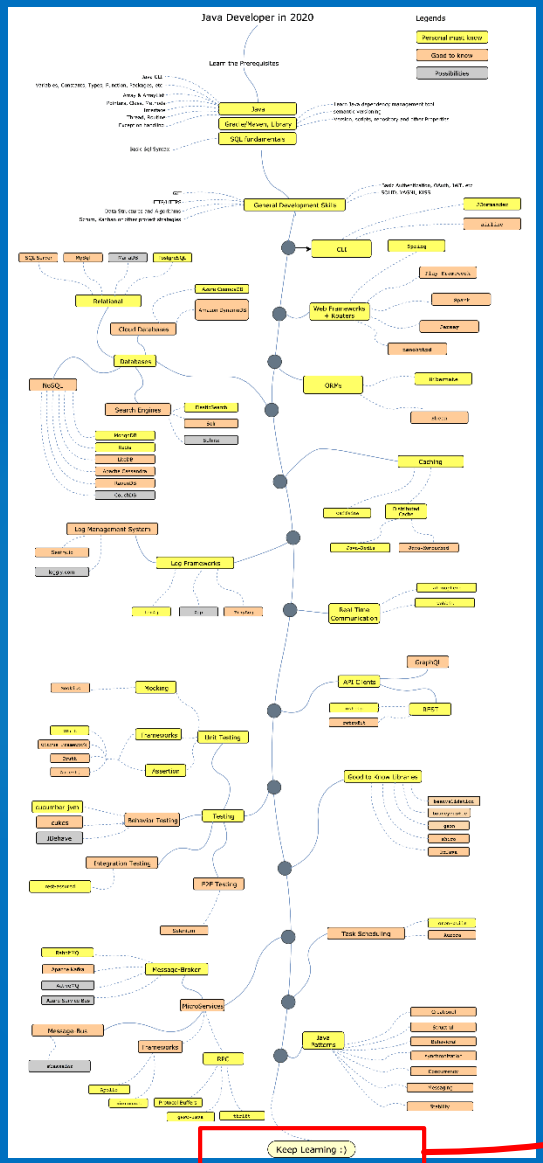


Online Judge System





Java Developer Roadmap





Keep Learning ☺

The Most
Important Thing